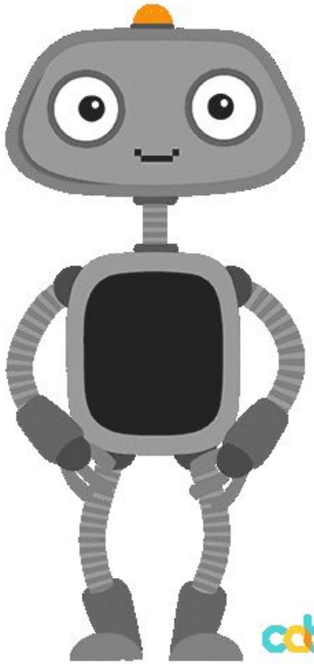
Week 09:
Design Project Kickoff

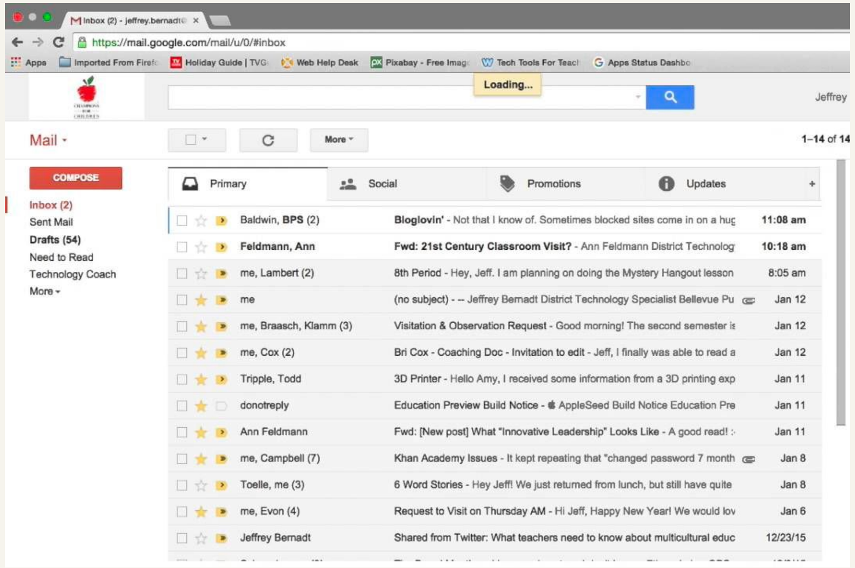
COMS 4170 • Spring 2026

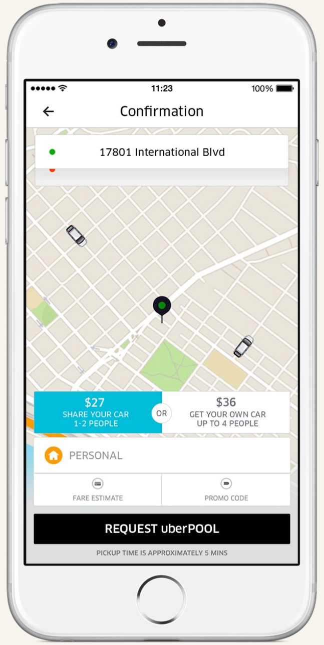
Prof. Celeste Layne
Tuesday March 24, 2026

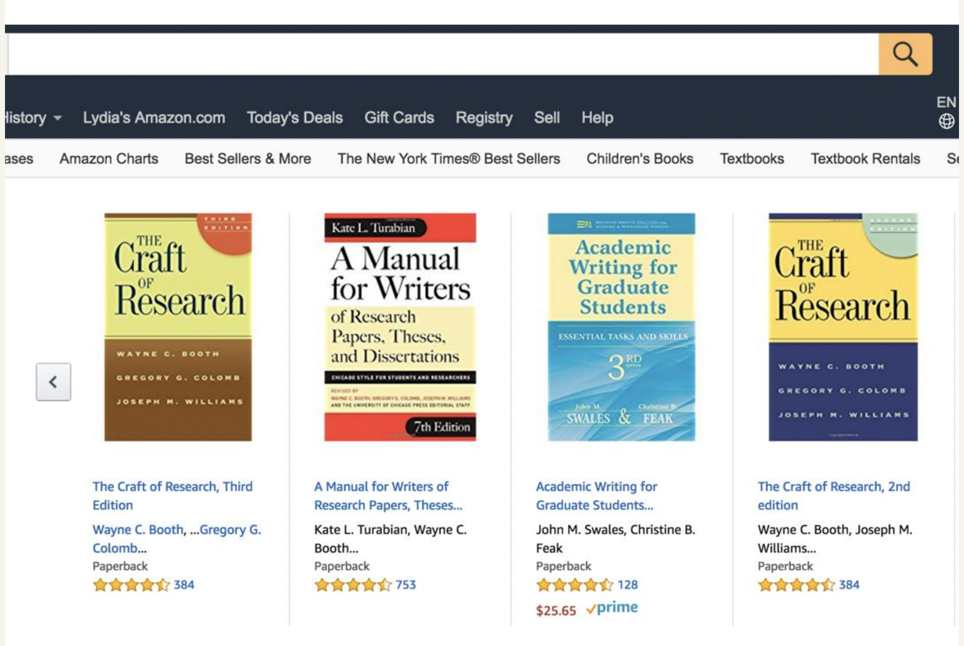


This midterm was a huge achievement ...

The main goal of many websites is to interact with data

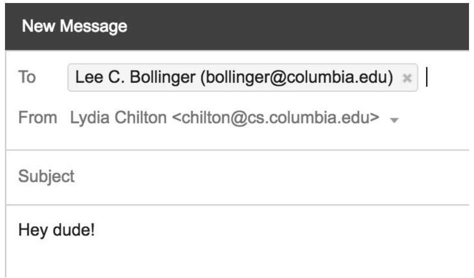


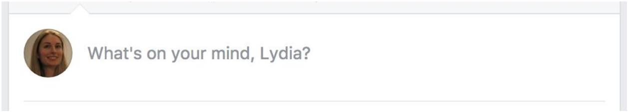


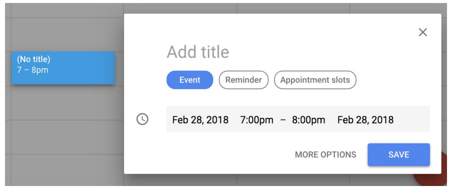


CRUD: Operations for interacting with a database

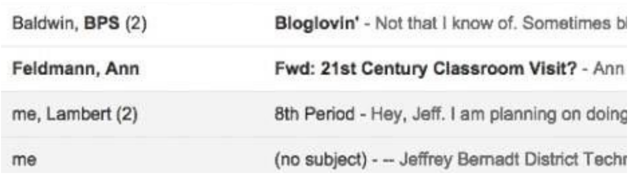
Create



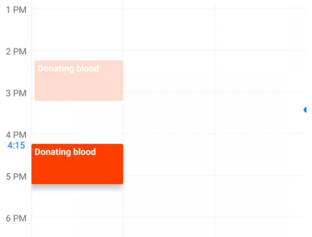




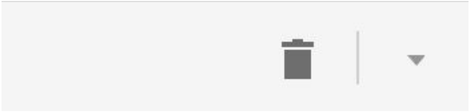
Read



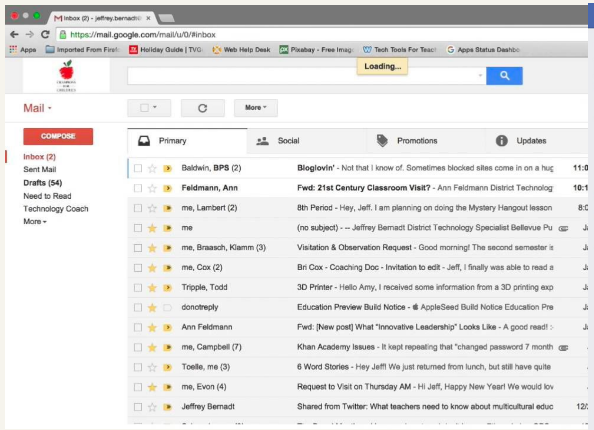
Update

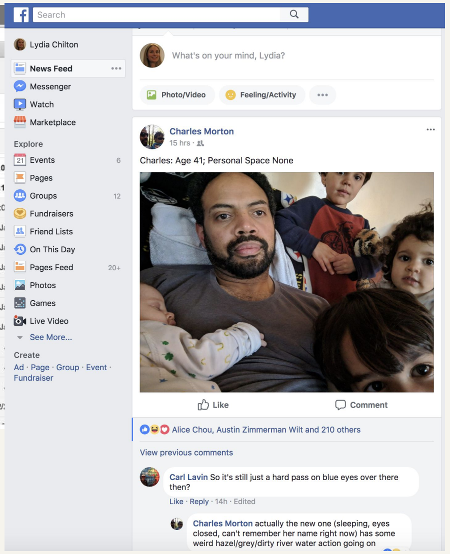


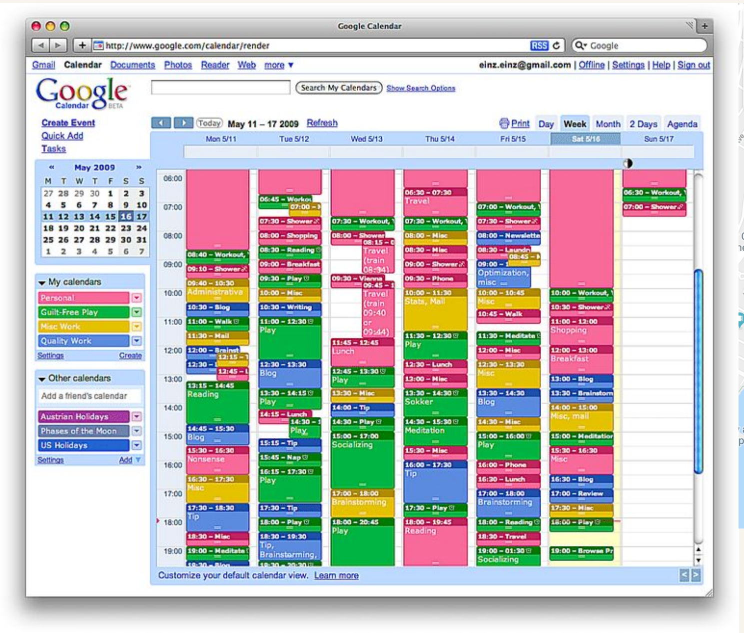
Delete

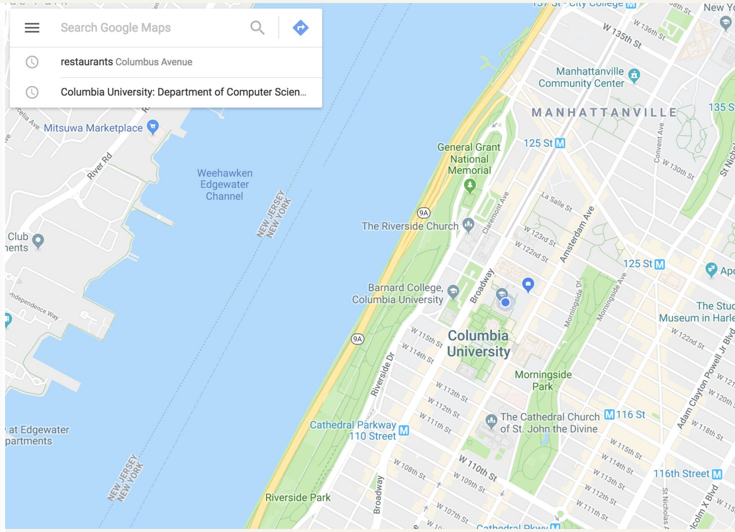


From a back-end perspective, database-backed websites are very similar;









However, they serve different data and different information needs.

Homework 7 Examples

127.0.0.1:5000

Apps | Your Projects - Ov... | UI Design | MLTD | MLTD piazza | NLP | Columbia Mail | Columbia | stance-internal | PyTorch documen... | transformers 2.3....

Castle Search Engine   Home   Create   Search for a castle or country!   Search

Our Mission:

Allow castle enthusiasts and prospective travelers to find and learn more about exciting castles around the world!

The Alcázar of Seville
Spain

Alcázar of Segovia
Spain

Moszna Castle
Poland

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Bon Appétit! Time to Dine

Home Create

Enter your restaurant search here | Search

Welcome! Feeling homesick? Hungry? Bored?
Peruse our selection of NYC-based French restaurants for pure deliciousness - satisfaction guaranteed!

10 Most Recent Entries:

Les Enfants de Boheme

Cozy, urban cafe with wall murals & locally sourced French bites, plus weekend brunch & happy hours.

LĒNA

Wine & Cocktail Bar - a cozy modern spot in LES. Dramatic décor, candlelit French music and a calm atmosphere take you from the hub of NYC to the hearth of nostalgic Basque region of France.

La Sirène-Soho

Casual bistro serving traditional Southern French cuisine made with seasonal ingredients.

The Playlist

Search | Submit | Create

The Playlist is a collection of rappers and their best songs; songs that are not just worthy of *our own* playlist, but worthy of *the* Playlist.

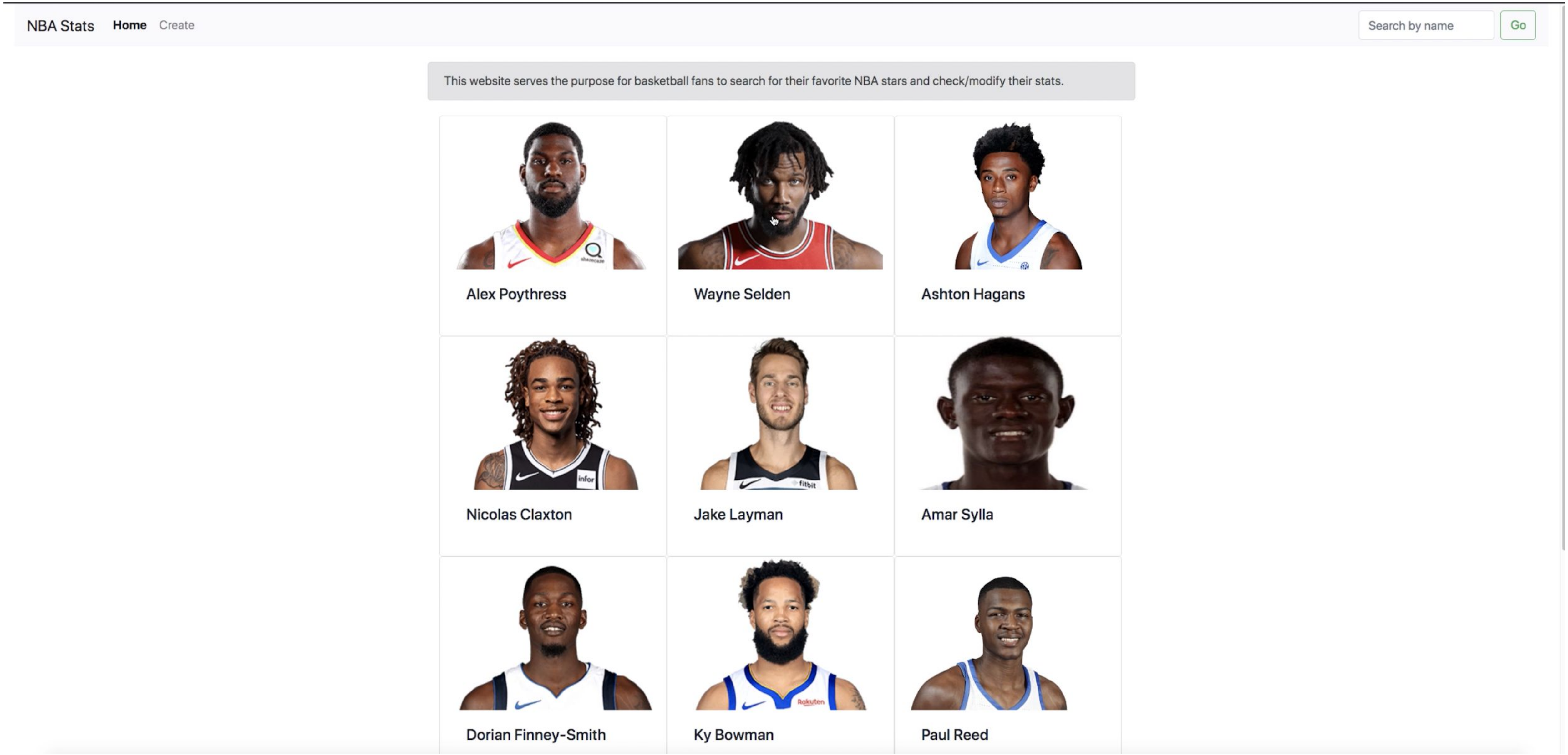
Kendrick Lamar

Cardi B

XXXTentacion

Ty Dolla Sign

J. Cole


NBA Stats
Home
Create
Search by name
Go
This website serves the purpose for basketball fans to search for their favorite NBA stars and check/modify their stats.
Alex Poythress
Wayne Selden
Ashton Hagans
Nicolas Claxton
Jake Layman
Amar Sylla
Dorian Finney-Smith
Ky Bowman
Paul Reed

localhost:5000

UFC Home Create

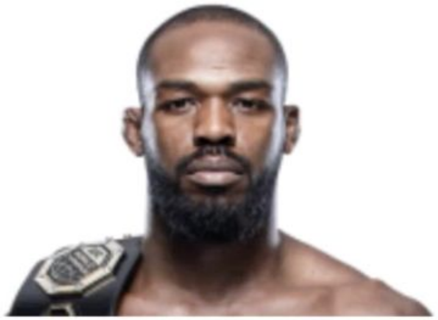
Search Search

Top Pound-for-Pound UFC Fighters

This platform allows UFC fans around the world to create a common source of information about their favourite fighters.

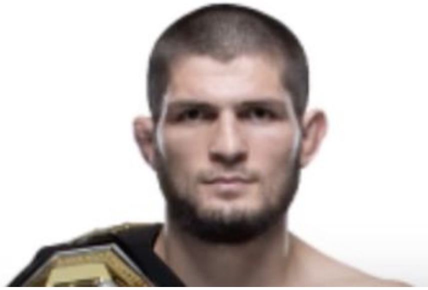
Conor McGregor Men's #9

Conor Anthony McGregor (born 14 July 1988) is an I...



Jon Jones Men's #1

At age 23, with his victory over Maurício Rua in 2...



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RUNNER'S QUEST | Add Trail | Search Trails | Go Running

Welcome to Runner's Quest! Become a member of our online community by picking a couple of trails you want to explore, and inviting your Runner's Quest teammates to experience them with you. We are your global team that is there to cheer you on.

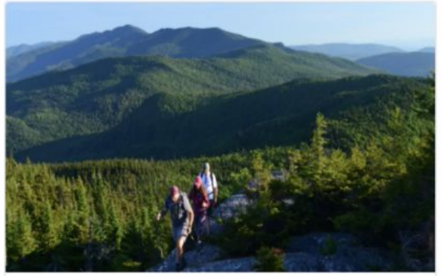
Shut-In Trail

Dale Ball Trails

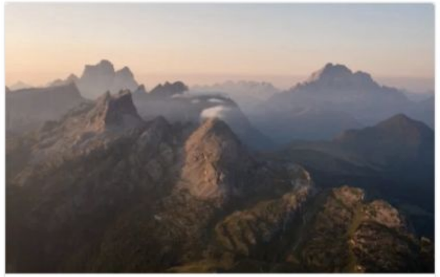
Colorado Trail

Palo Duro Canyon State Park

El Moro Canyon Loop



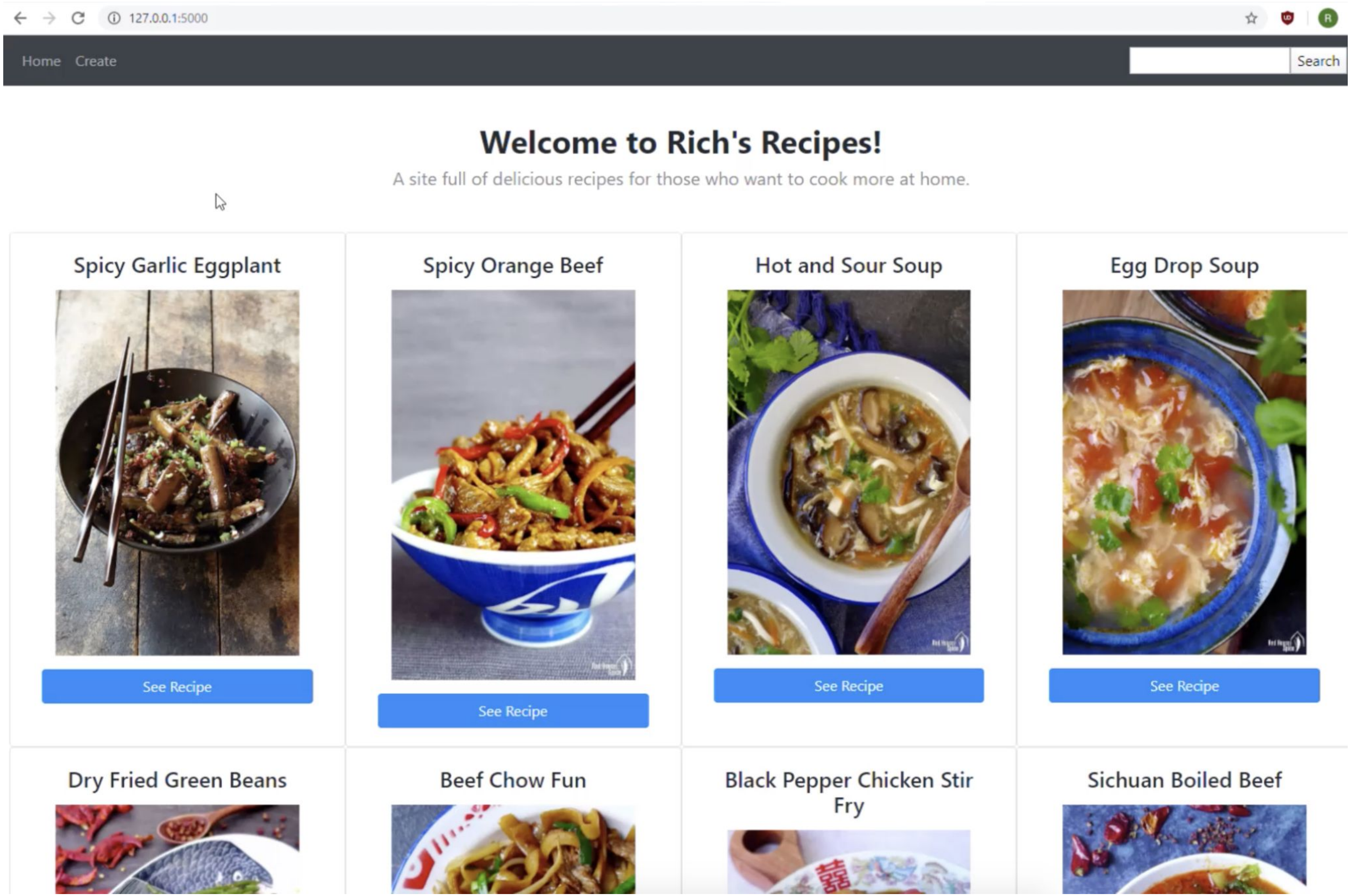
The Long Trail



Alta Via 1

Teton Crest Trail

Prenj Massif to Vran Mountain on the Via Dinarica


127.0.0.1:5000
Home
Create
Search
Welcome to Rich's Recipes!
A site full of delicious recipes for those who want to cook more at home.
Spicy Garlic Eggplant
See Recipe
Spicy Orange Beef
See Recipe
Hot and Sour Soup
See Recipe
Egg Drop Soup
See Recipe
Dry Fried Green Beans
Beef Chow Fun
Black Pepper Chicken Stir Fry
Sichuan Boiled Beef

Jazz Archives Home Create

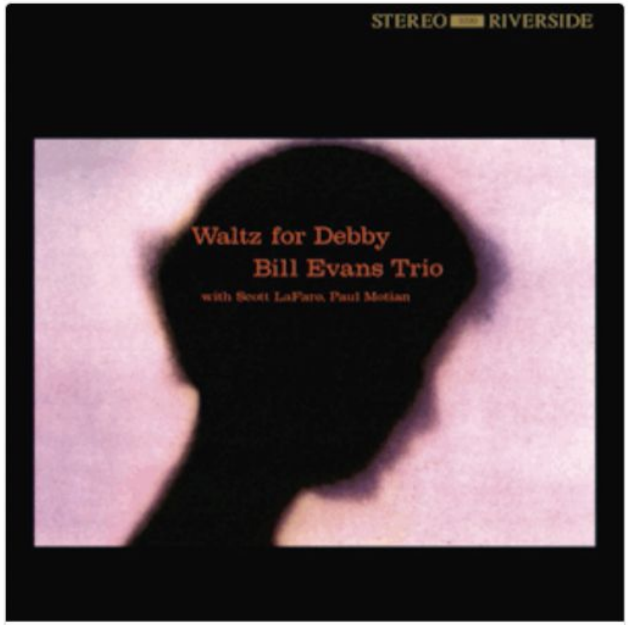
Titles, Artists, Songs, Musicians... Search

Welcome to the Jazz Archives database

Here you can explore classical jazz albums using our friendly search bar or just diving into our most recent additions.

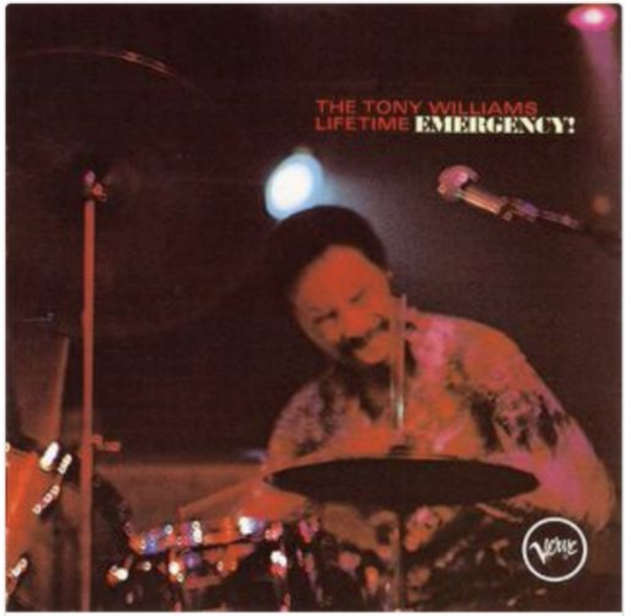
Did you hear a **jazz record** and want to know more?
Do you like a **musician** and want to know other records where they play?
Do you want to start listening to jazz and **don't know where to start**?

You've come to the right place!



Waltz For Debby (1962)

Bill Evans Trio



Emergency! (1969)

The Tony Williams Lifetime

It Could Happen To You (1958)

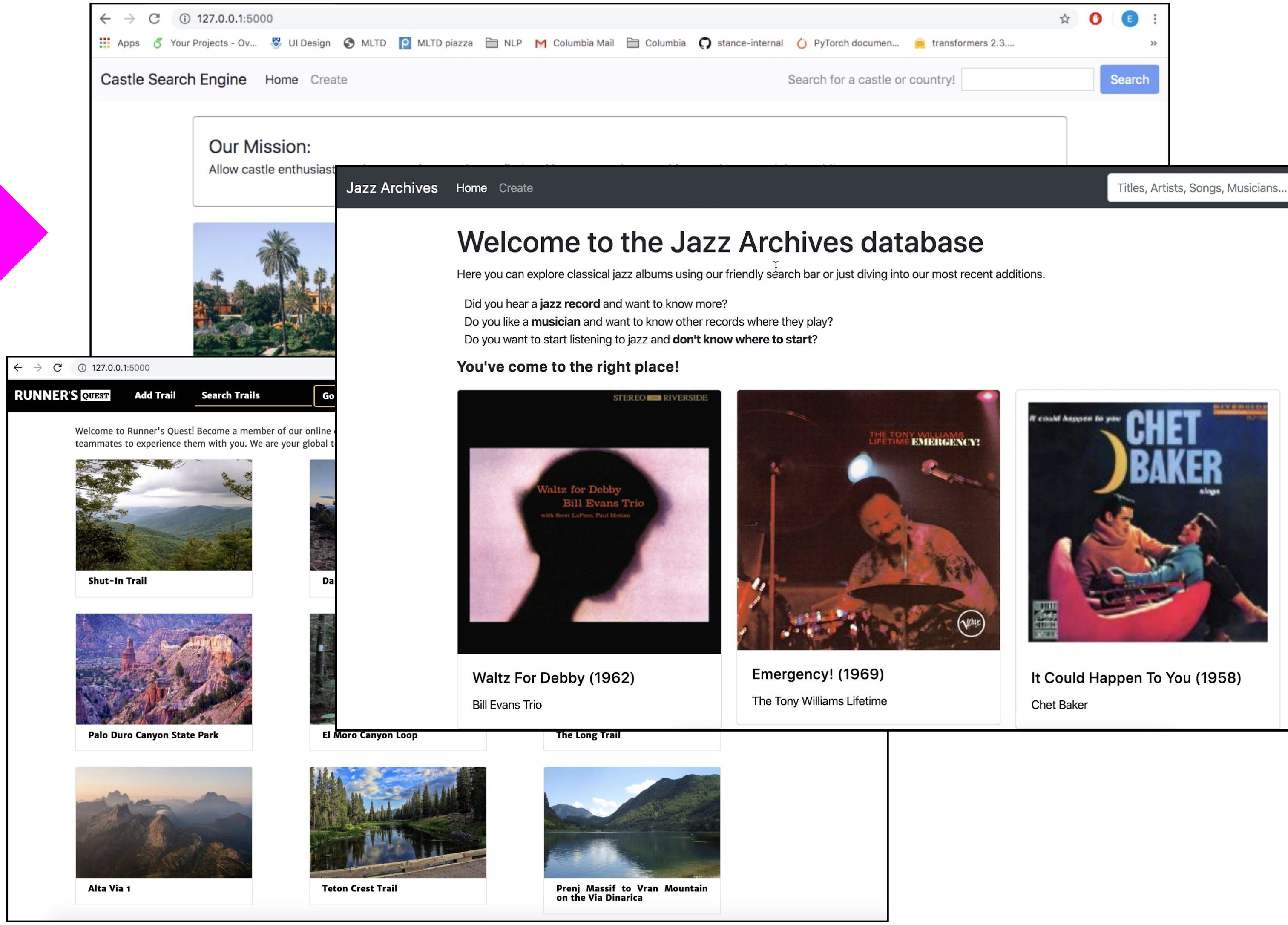
Chet Baker

You are now a **user interface programmer**

Given specifications, you can create user interactions

Part 1 – Usable Functionality:

1. **Menu/Navigation**.
   a. For consistency, all the templates should be rendered with a shared template that contains a navbar.
   b. The navbar should contain:
      i. A home link (at the "/" route)
      ii. A text box to enter a search query and a "go" button (at the "/search" route). When the user presses enter on the search bar it should also "go".
      iii. A create link (at the "/create" route)
2. **Home**. The home link should render at the "/".
   a. It should contain a one sentence summary of the mission of the site. This mission should make it clear who the intended user is and what specific goal it helps them achieve.
   b. It should show the latest 10 entries added to the database to entice the viewer to click on something and start exploring.
   c. Each of the 10 entries should be formatted as a Bootstrap Card that contains an image and the title of the item. If there is some other essential field, it can show that too, but it should not show all the data fields – it's meant to be a summary.
   d. When you click the image, it should take you to the page for viewing the item.
3. **Search**. When the user presses "go" on the search link (or presses enter), it should search for the items and return a list of all matching results.
   a. **Flexibility**. The query must do substring matching that is not case sensitive on the title and one other text field.
   b. **Feedback**. In addition to returning the results, the page must say how many results there are. If there are zero results, you don't need to do anything other than say there are zero results.
   c. **Feedback**. When you present the results to the user, the bit that matches the substring must be easy to scan for, according to gestalt principles.
4. **State/Options/Transitions**. On the template for creating a new database item, you will still have input boxes for all the fields the user must input. In addition:
   a. **Error Detection**. When creating a new database entry, there must be error handling on all the fields. If the field must be a number, then ensure it is a number. At the very least, you can check that the field in not blank (remember to trim the text to test if it's blank). Design the error feedback so that it directs the user's attention to the right place to correct the error.
   b. **Transitions**. After the user presses "submit" and the data successfully submits, allow the user to either view the item or enter a new item.
      i. At the top of the page it should say, "New item successfully created." With a button or link that says "see it here" (or words to that effect). This links to a page for viewing the item.
      ii. Additionally, the input boxes should clear and the focus should be placed on the first text box so the user is ready to submit another item.
5. **State/Options/Transitions**. There will no longer be a separate /edit/<id> route. Editing will now be done in /view/<id>



The next step is to become a user interface designer

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   c. **Feedback**. When you present the results to the user, the bit that matches the substring must be easy to scan for, according to gestalt principles.
4. **State/Options/Transitions**. On the [illegible] database item, you will still have input boxes for all the fields the user must input. In addition:
   a. **Error Detection**. When creating a new database entry, there must be error handling on all the fields. If the field must be a number, then ensure it is a number. At the very least, you can check that the field in not blank (remember to trim the text to test if it's blank). Design the error feedback so that it directs the user's attention to the right place to correct the error.
   b. **Transition**. After the user presses [illegible] the data successfully submits, allow the user to [illegible] view the item or enter a new item.
      i. At the top of the page it should say, "New item successfully created." With a button or link that says "see it here" (or words to that effect). This links to a page for viewing the item.
      ii. Additionally, the input boxes should clear and the focus should be placed on the first text box so the user is ready to submit another item.
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You identify the user

You identify the problem

You come up with a solution

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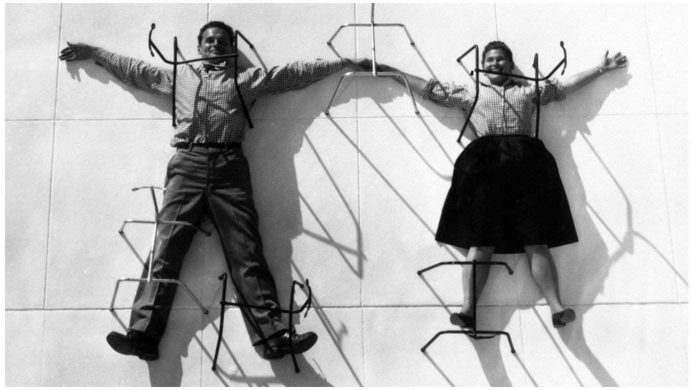
What is Design?

"Design is a plan for arranging elements to accomplish a particular purpose."

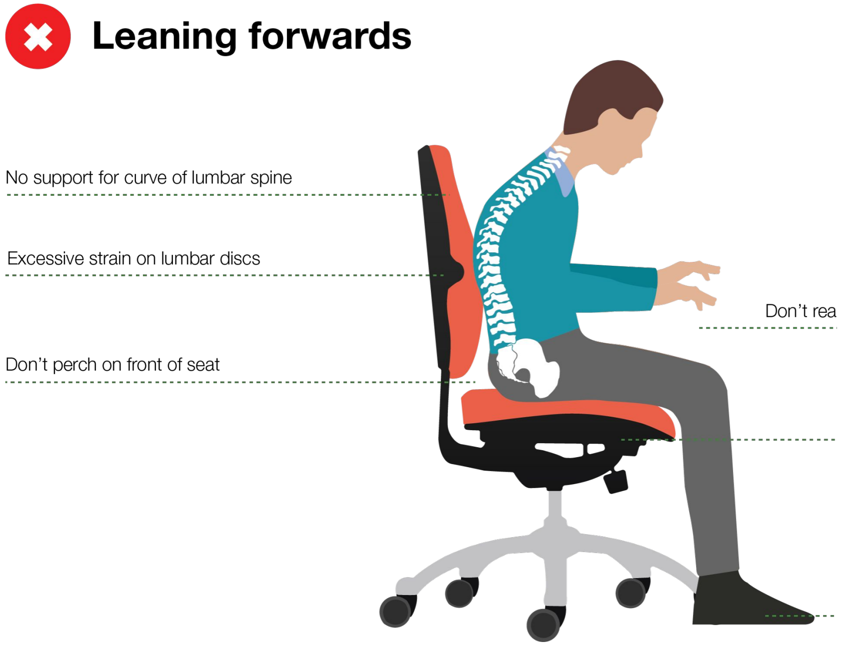
– Charles Eames, designer, architect and film-maker



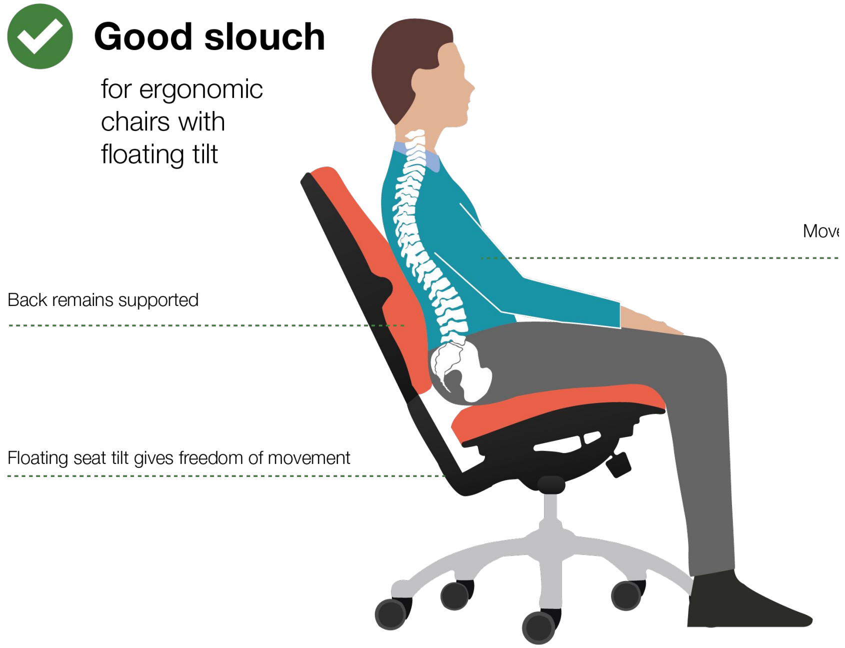
Design is **not some magical leap** where a brilliant idea comes from outer space.

Design is an iterative progress where you work with users to identify and solve their problems

Sitting all day hurts.



Why does it hurt?
How do people sit?



What does good sitting look like?

For people who sit all day in an office, alleviate back pain by designing a chair that supports the lower back

Identify users needs:

Test solutions on users:

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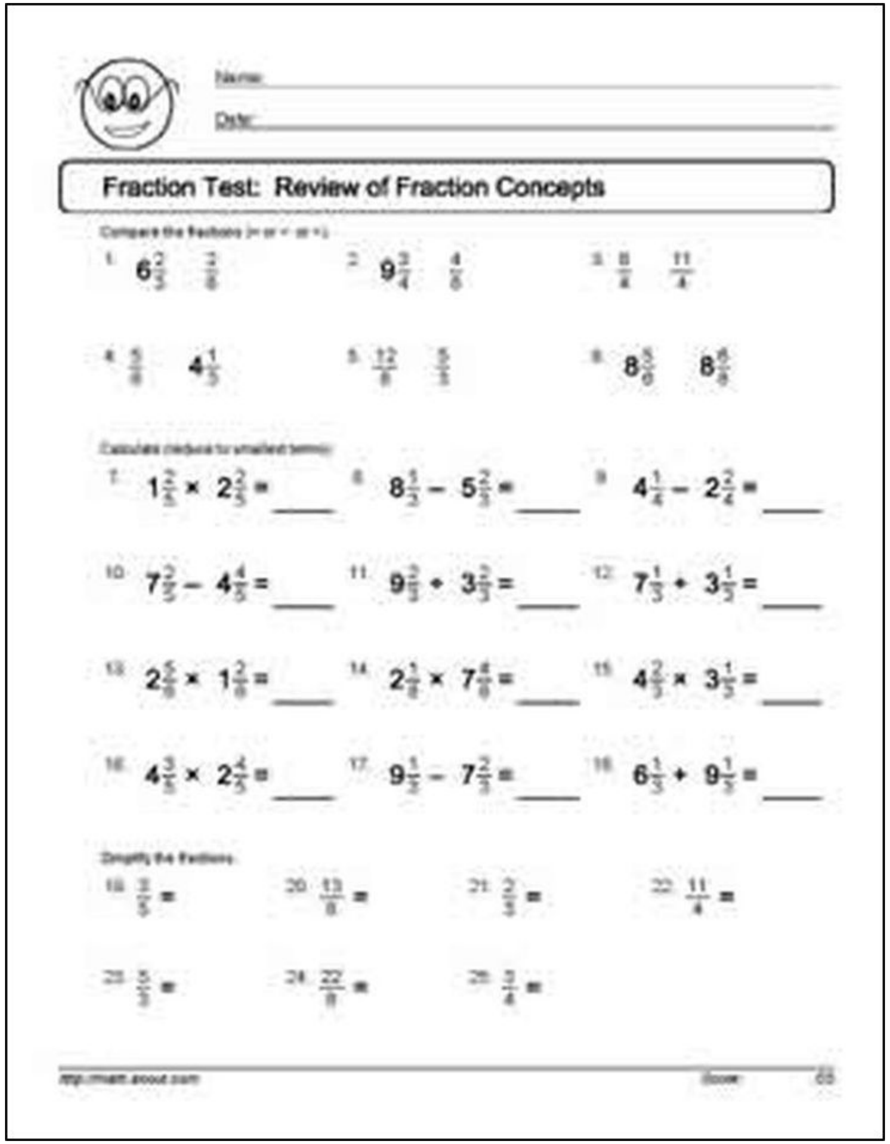
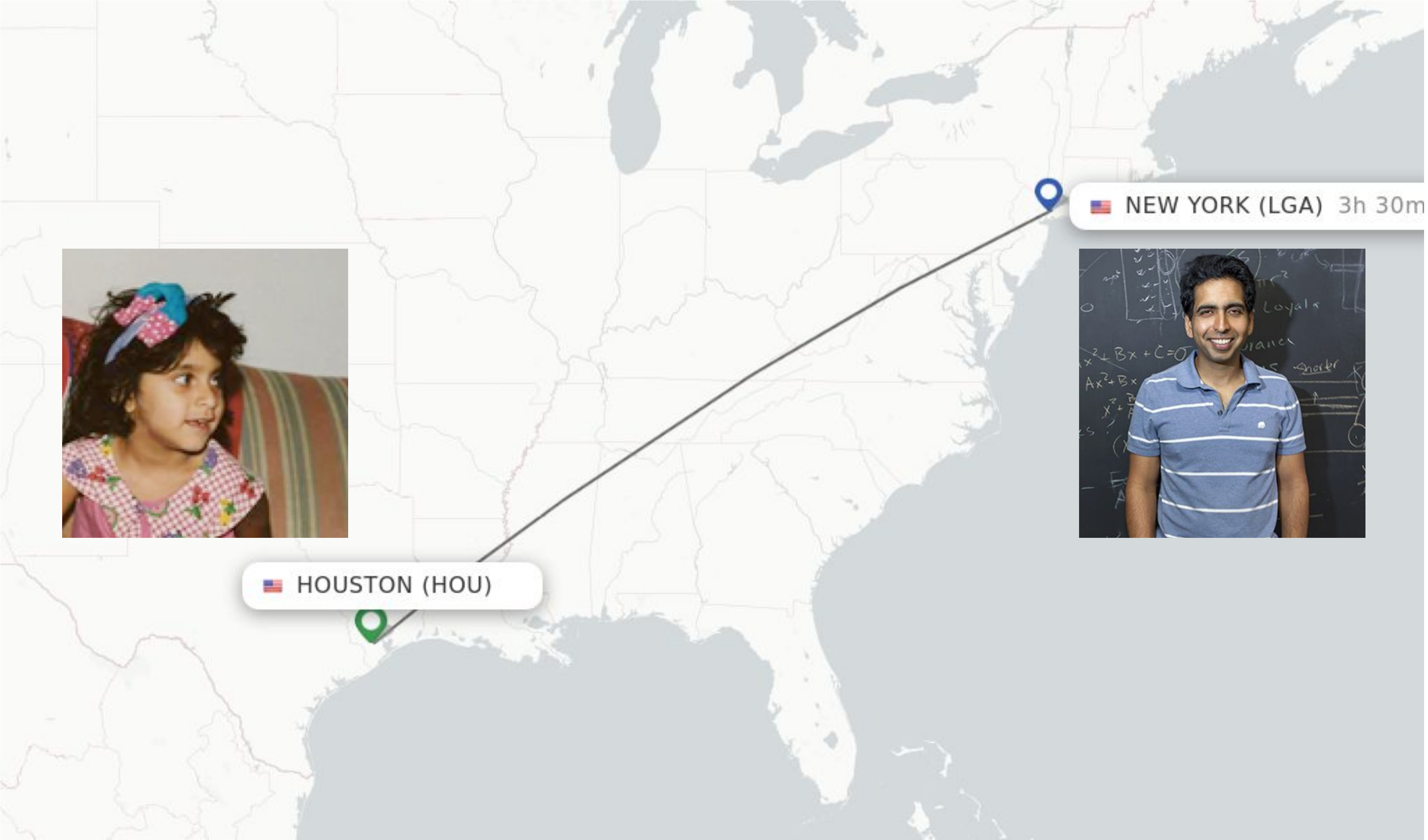
Let me tell you a story

This is Nadia. She is 11 years old. She lived in Houston.

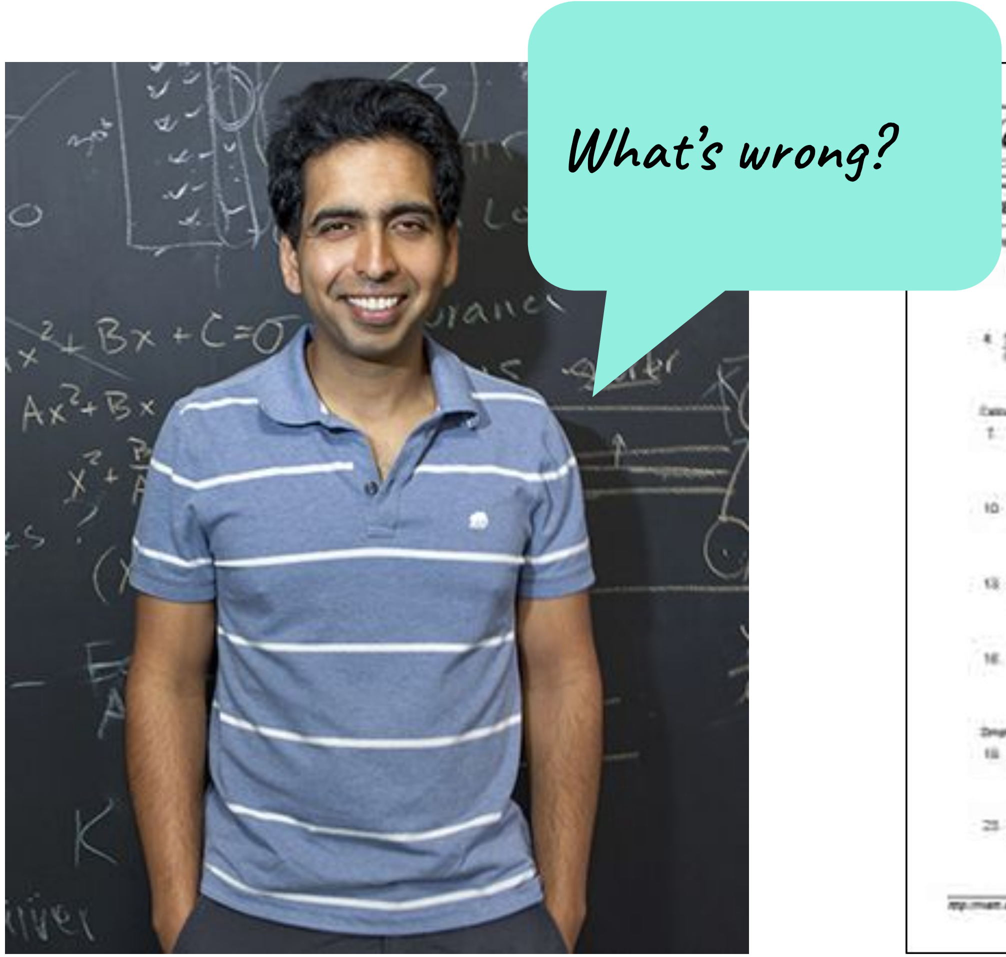
Nadia was struggling with fractions at school

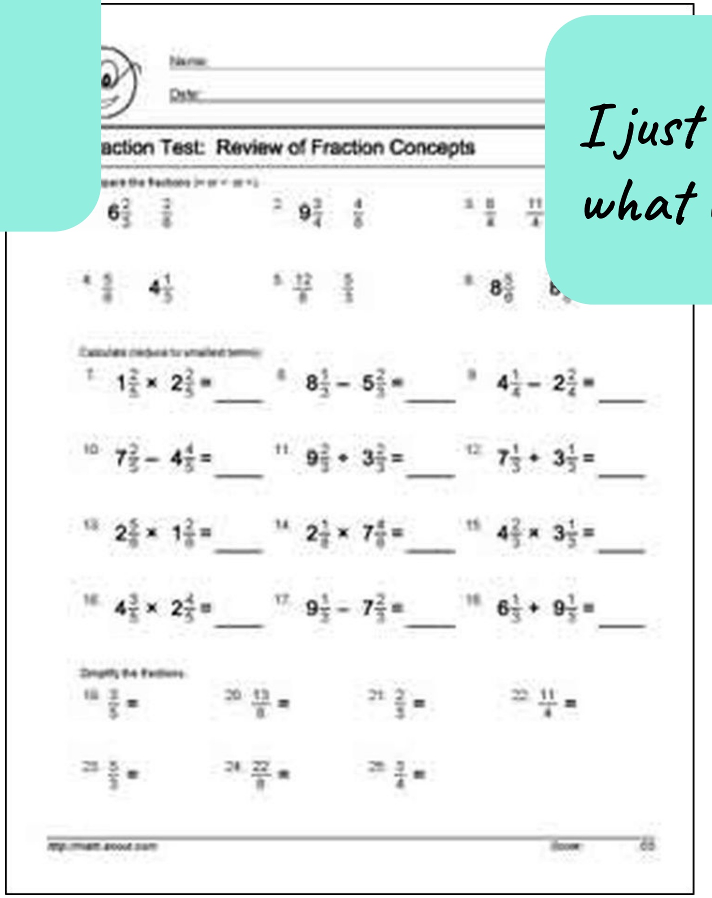

Name:
Date:
Fraction Test: Review of Fraction Concepts

Her uncle wanted to help; but he lived in New York City



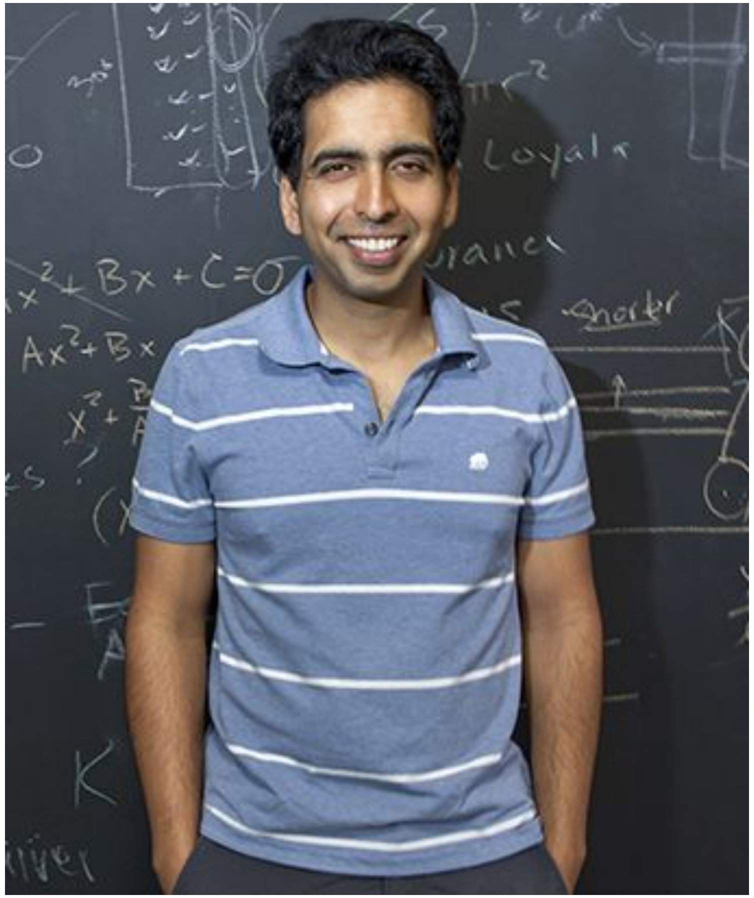
He asks about her fractions homework

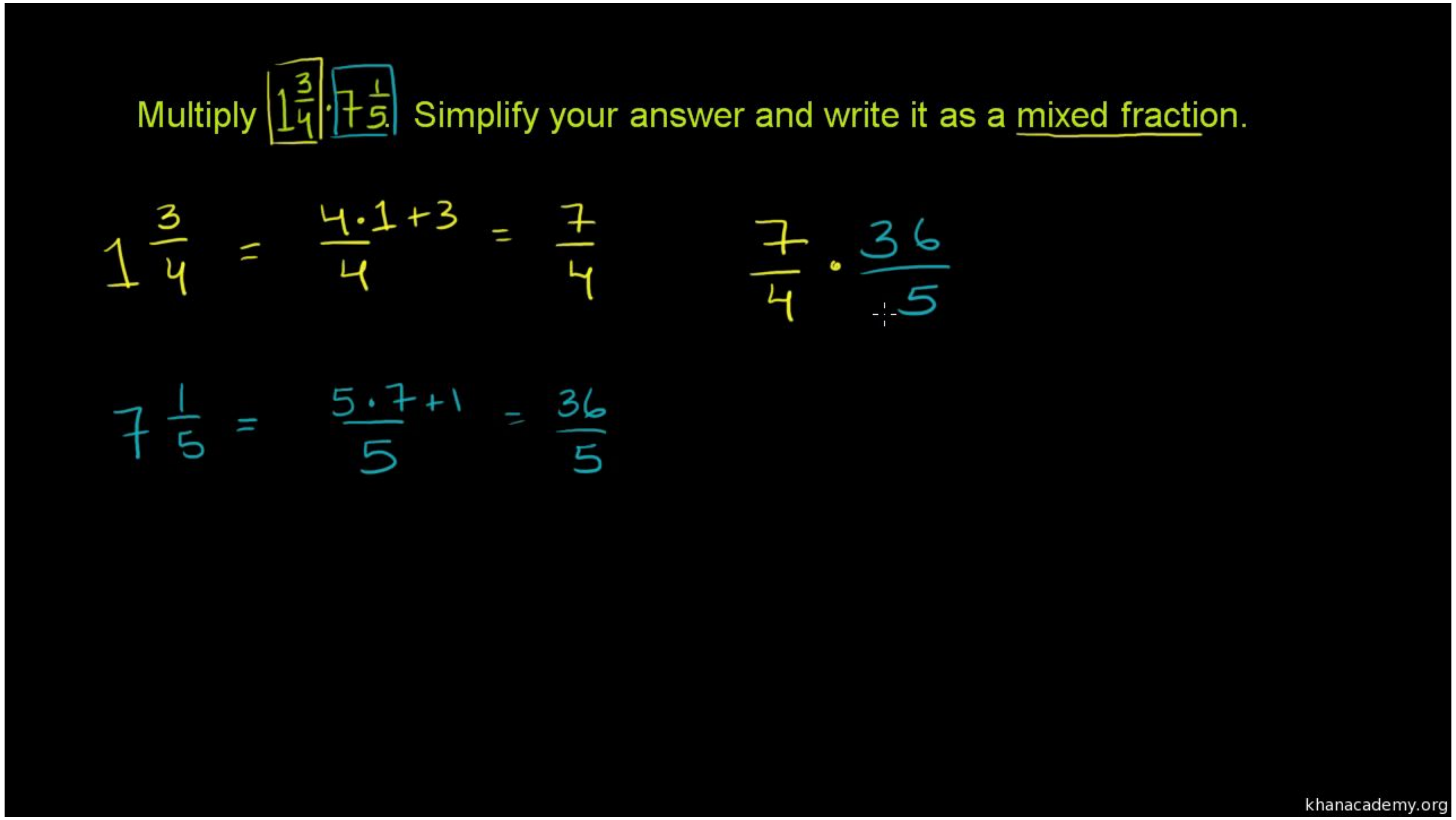




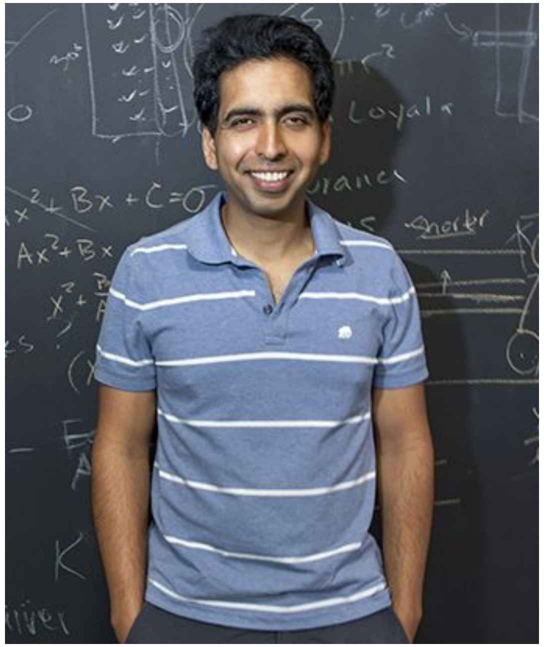
He has the insight that if she saw more examples, she could figure out how to solve fractions.

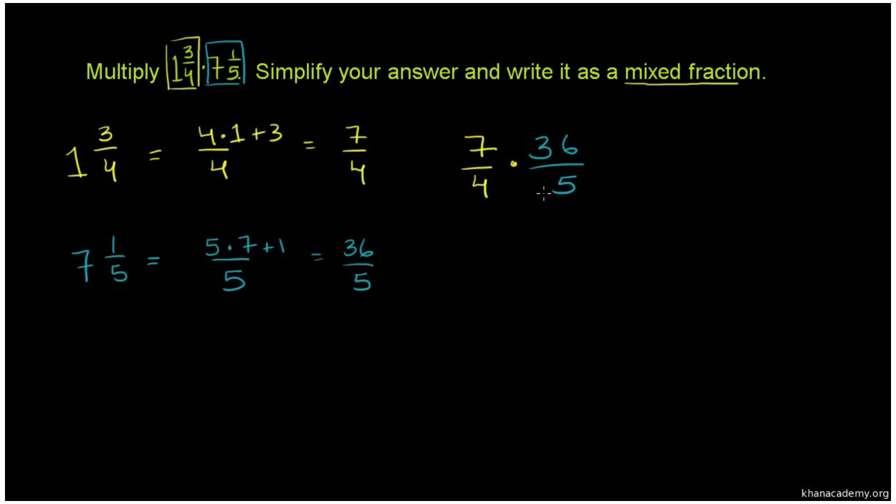
He makes videos and uploads them to YouTube

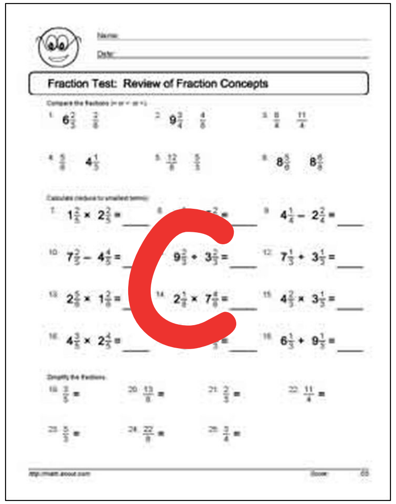




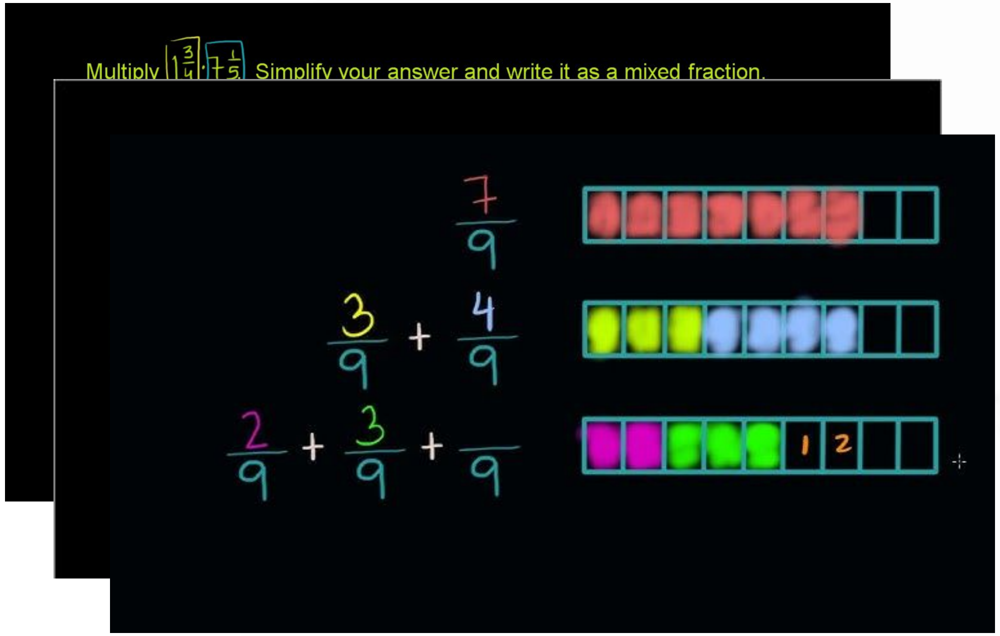
…and he shows it to Nadia to see if it helps.

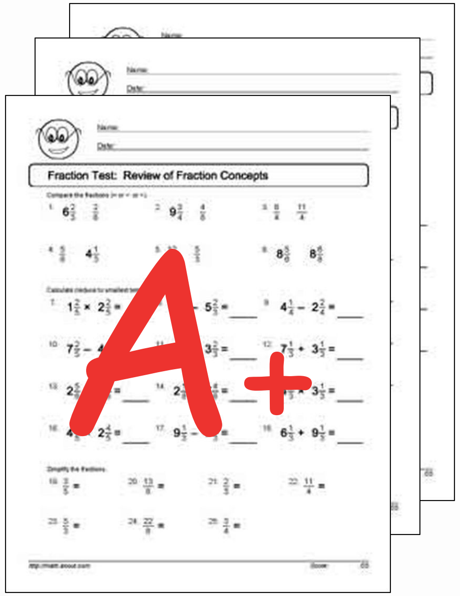


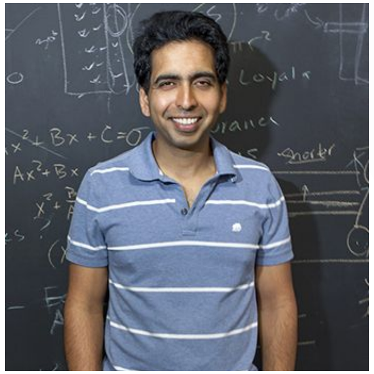




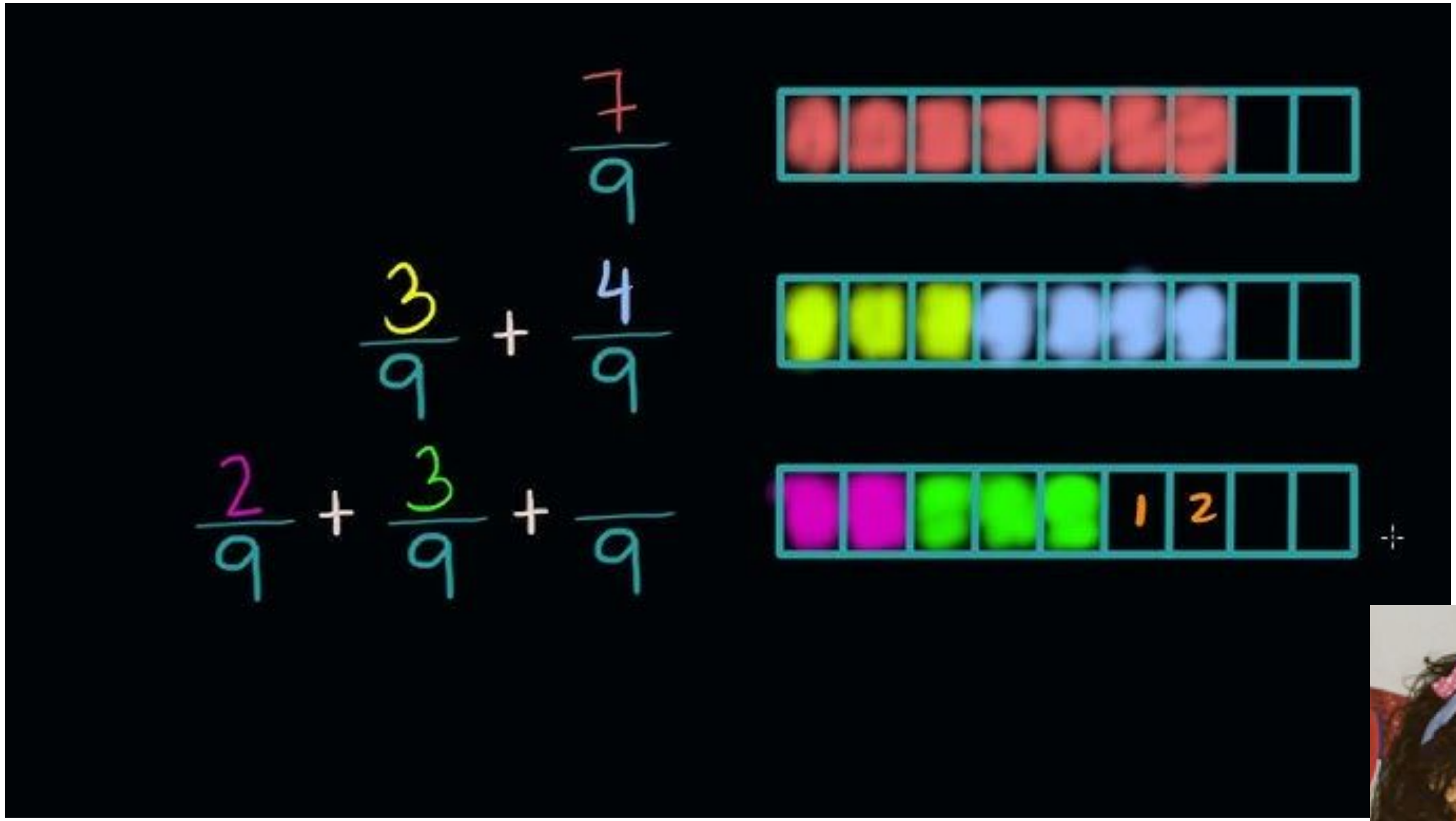
...and he improves it again and again

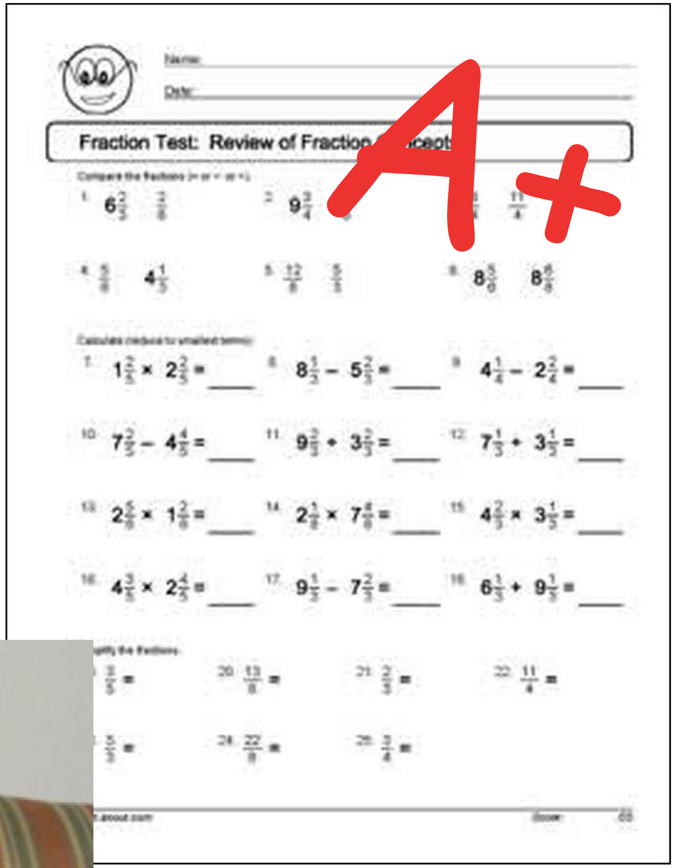




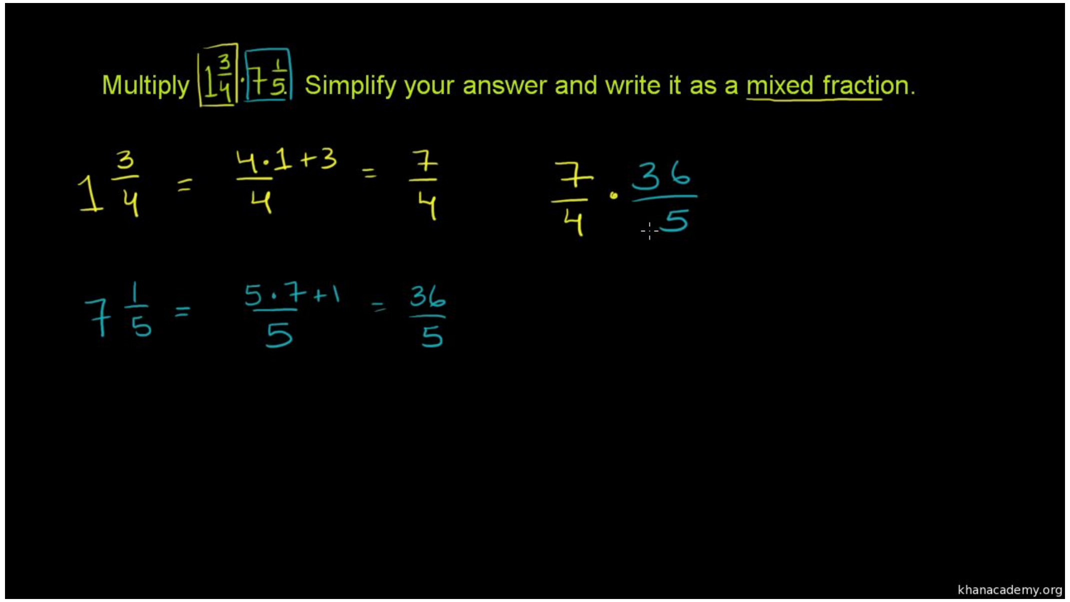


This video helps Nadia. She aces her fractions test.

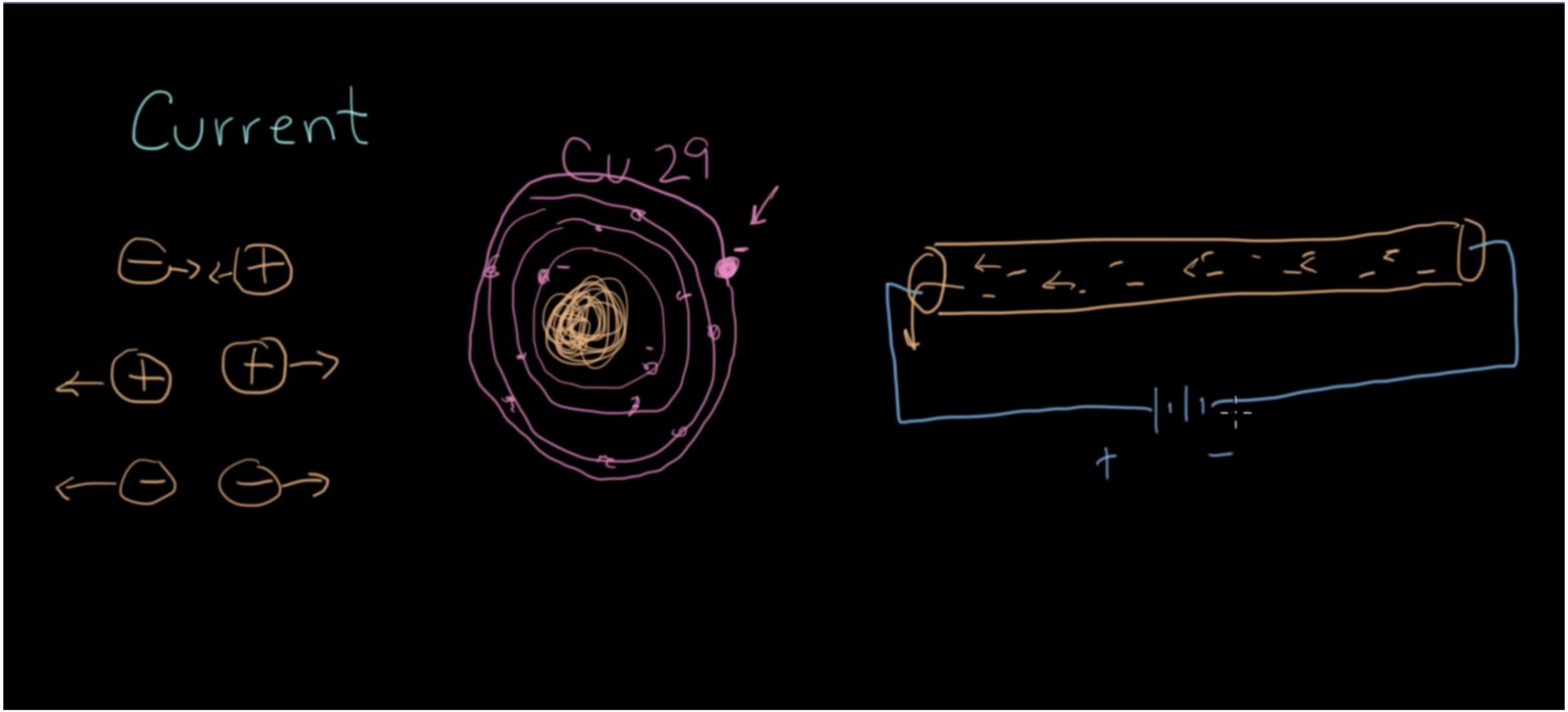




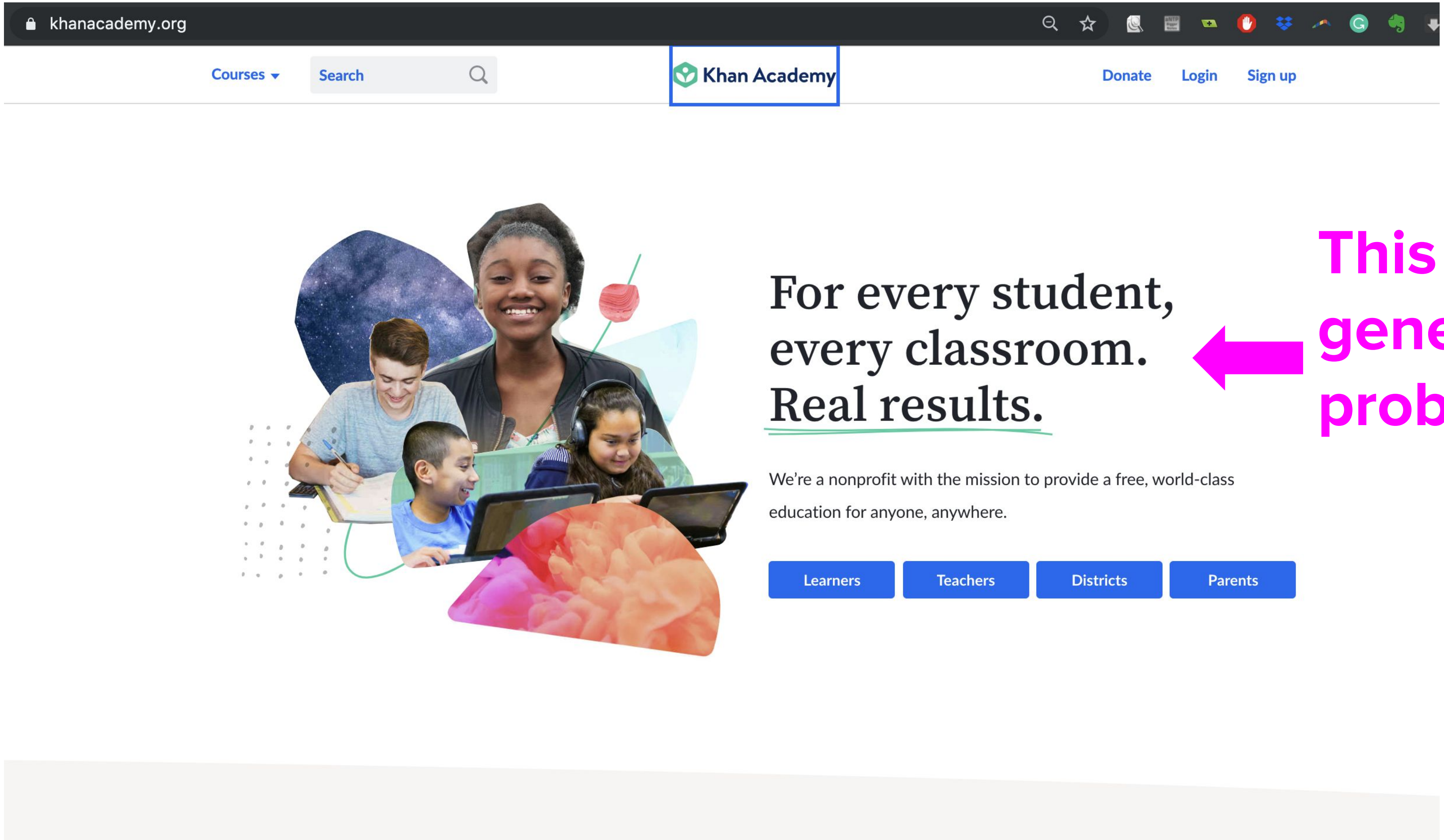
The videos help other students learn their fractions, too



Videos in this format help students learn other subjects

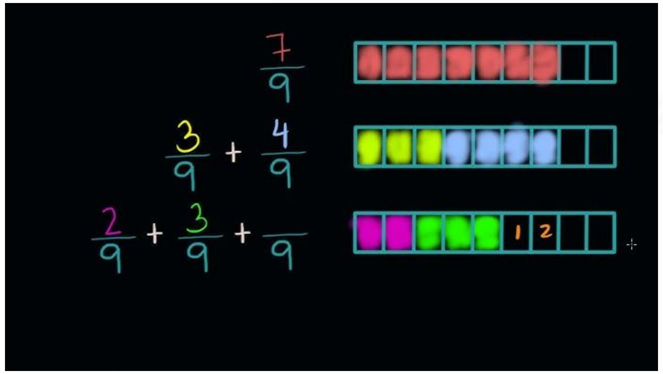


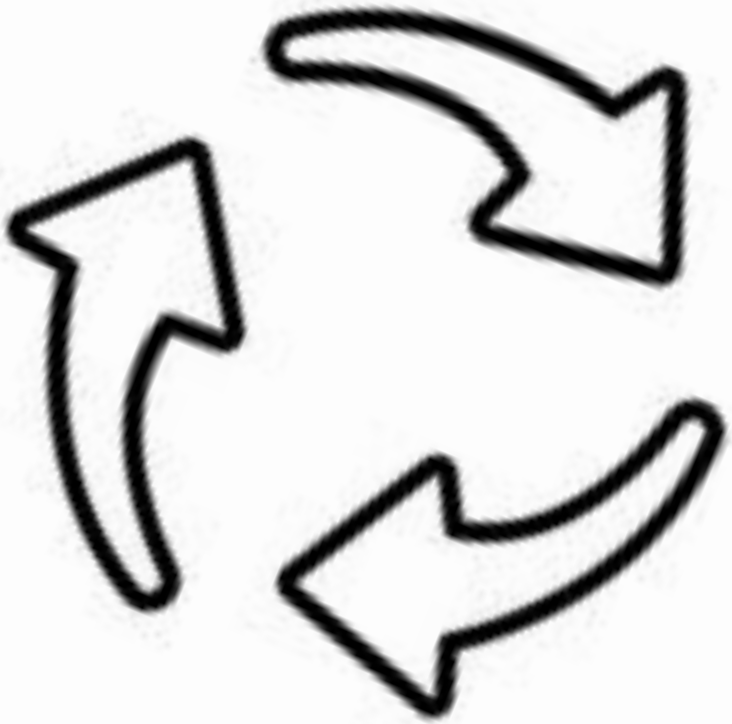
Those videos became Khan Academy

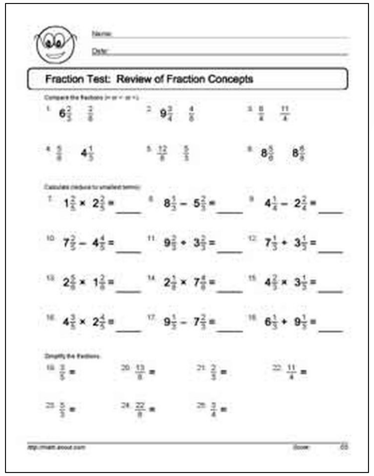


This is a general problem

However, it started by helping a specific person with a specific problem, and a lot of iteration.







Don’t start big. Start small.

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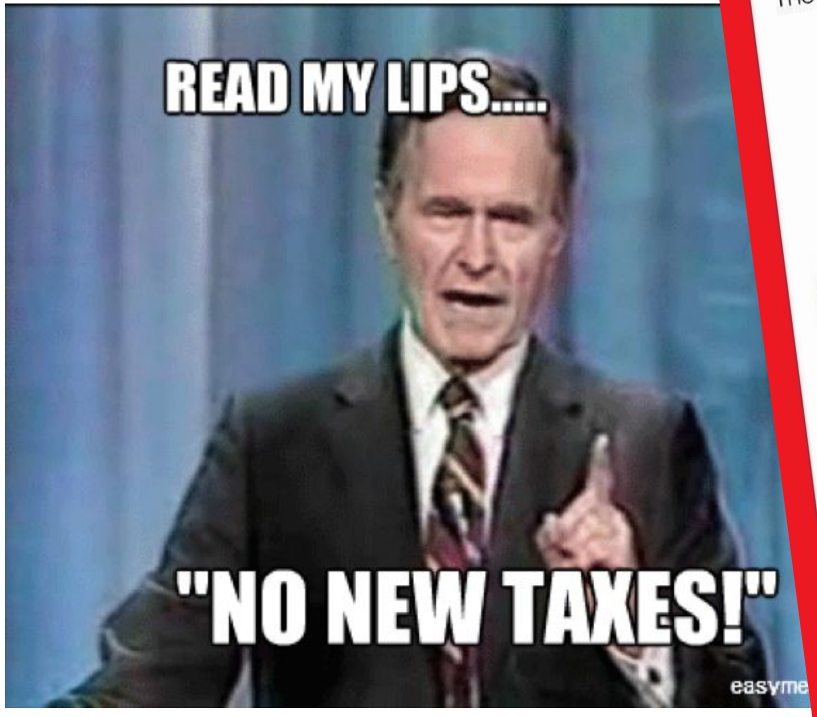
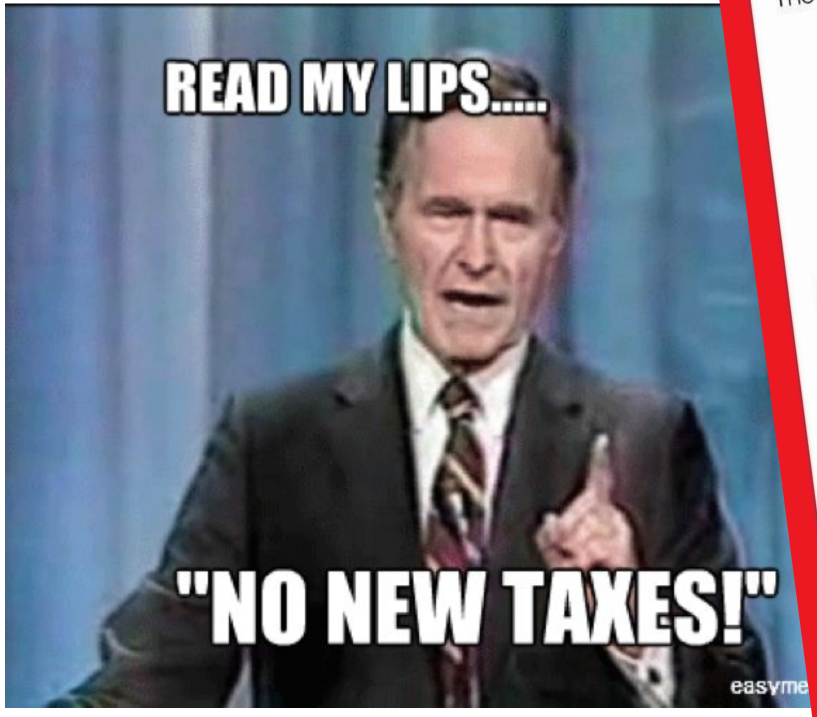
Specific vs. General Needs

What are the kinds of things politicians, beauty queens and Silicon Valley say they will solve?

In the media, people often claim they are solving big problems.



There is something appealing about how grand a big problem sounds; but…



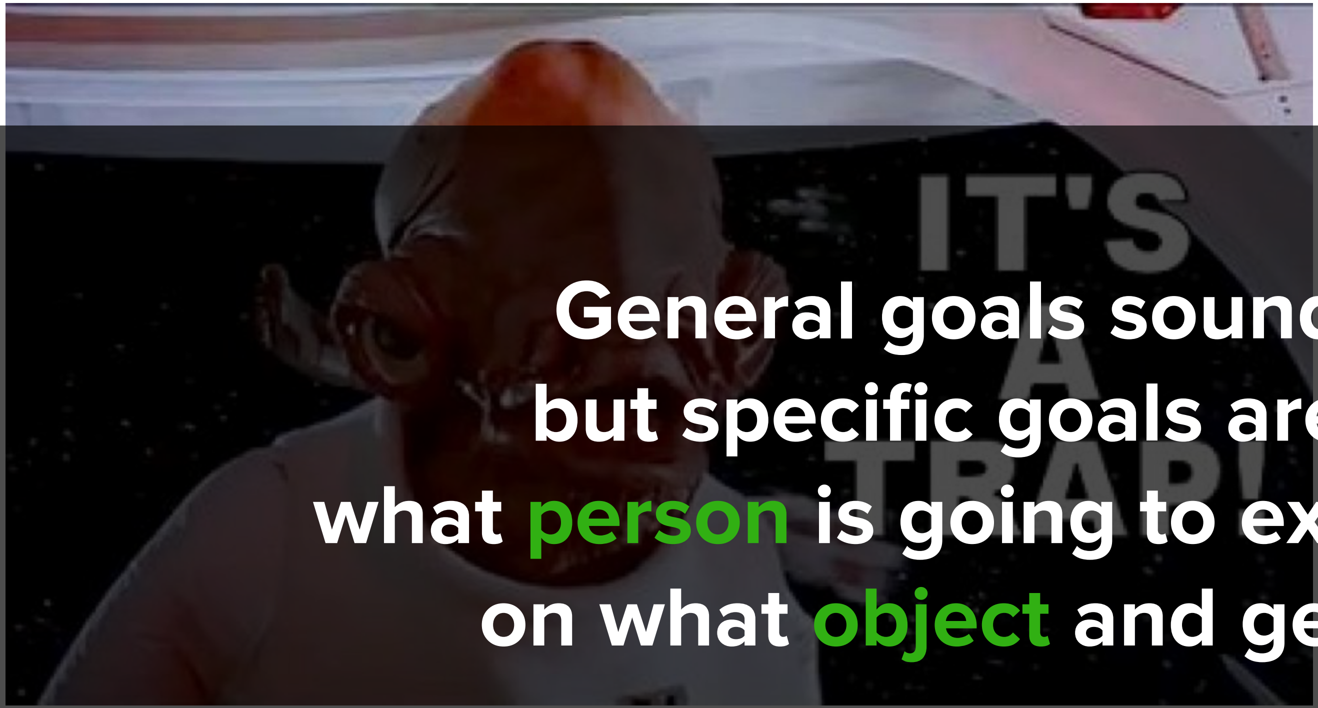
What is the problem with general goals?

They are not actionable.

General vs. Specific goals

Goal 1: Clean the house

Goal 2: Fold the basket of laundry



General goals sound appealing;
but specific goals are actionable:
what person is going to execute what action
on what object and get what value?

General goals are actually domains

Domain:

“Clean the house”

Specific goal:

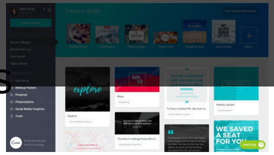
“Fold the basket of laundry”

What is the risk with a specific goal?

“Fold the basket of laundry”

Specific goals can be trivial.

However, if you start specific you can usually generalize.

	Domain	Specific Need	Generalized to
amazon	Online shopping	Uncommon books	Clothes, food, Amazon Fresh, other sellers 
facebook	Social networking	Harvard students looking up dorm, classes, relationship status	Ivy League US colleges Everybody 
Gmail	Read or send emails	No page reload Never delete	Chat, GDrive
Canva	Graphic Design for novices	High school yearbooks	Posters, instagram posts 

If you start specific, you can usually generalize.

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Design Project

Design Challenge

- Design and build a **web** application
- To help a user learn an introductory topic **interactively**
- Within a **domain of your choosing**
- The interactive experience must center around **media (images, video, audio)**
- The interactive experience should help users assess their learning with a quiz
- Users should keep learning through feedback from the quiz
- It should take under 10 minutes **total**

Design Challenge

The USER is another student in this class

Why?

- So that you can test your designs on actual people
- Your TA must think that this is something valuable for them to learn
- Consequently, you cannot design for any of the following:
  - Children or teenagers who are not in this class
  - People who only speak Serbian (everyone here speaks English)
  - Majors not represented in this class
  - A topic your TA is uninterested in (your TA has to like the idea)

Design Challenge

The TOPIC must be in a domain of your choosing; but it must be focused enough to teach in 10 minutes.

- Examples of broad domains:
    - chess, basketball, art history, music etc
- Examples of focused topics:
    - Chess: How and when to perform 3 different opening moves in chess for chess beginners
    - Basketball: How to run a pick and roll in basketball for casual NBA fans
    - Art History: How to tell Impressionist paintings from Post-Impressionist paintings for ArtHum students
    - Music: How to mix a drop swap for aspiring DJs

Design Challenge

Project Logistics

- **Weekly homework** will build up to the final project (5% of grade)
- Final submission is worth **20% of your grade**
- This project is to be completed **individually**
  - You will meet with a TA to receive feedback
  - You will meet with your TA in a group to get feedback and support each other

Design Insight for Teaching:

People learn through interaction and feedback, not from reading long streams of text.

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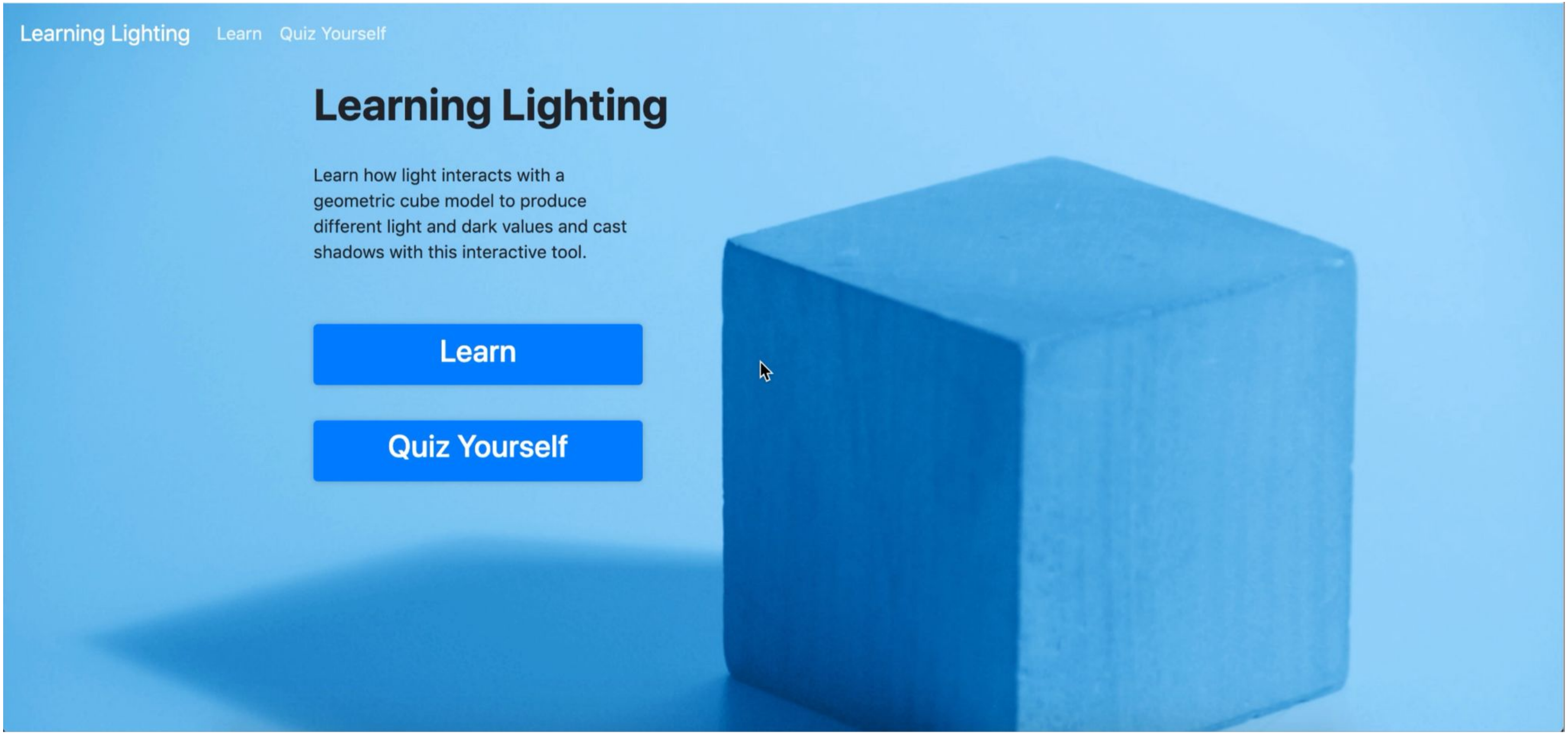
Examples


Learning Lighting
Learn
Quiz Yourself
Learning Lighting
Learn how light interacts with a geometric cube model to produce different light and dark values and cast shadows with this interactive tool.
Learn
Quiz Yourself

How does it fulfill the requirements?

Domain	Lighting
Topic	Lighting from five directions
Media	3D model
Interaction	Click the model
User	Would you learn something from this?

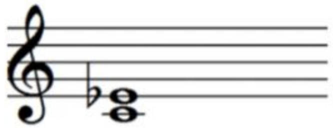
Music Interval Training Major 2nd Minor 3rd Major 3rd Perfect 4th Perfect 5th Major 6th Perfect Oct. Quiz

< prev. interval

Minor Third

next interval >

Step 1: Listen



A Minor Third has 3 half steps between the lower note and the higher note. Try counting the number of lines and spaces between the two notes on the image above.

The "minor" quality indicates that this interval comes from the minor scale of its lower note, and that the interval sounds slightly dissonant or unhappy.

0:00 / 0:06

Step: 1 2 3

How does it fulfill the requirements?

Domain	Music
Topic	Identifying intervals
Media	Audio recording and playable piano
Interaction	Play music; press keys on the piano
User	Would you learn something from this?

ASL for Kids! Learn Mode Quiz Mode

Click around the classroom to learn the signs!

Hover your mouse over the classroom to see where to click.

Click to see the sign!

You have 11 items left to learn!

How does it fulfill the requirements?

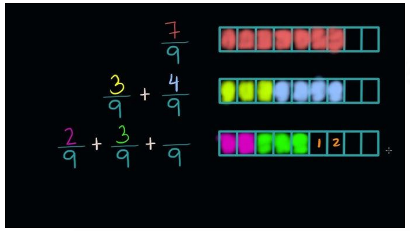
Domain	Sign Language
Topic	Learning objects in a classroom
Media	Video (for sign language), interactive images
Interaction	Click on objects in image to learn the names
User	Beginner sign language learner

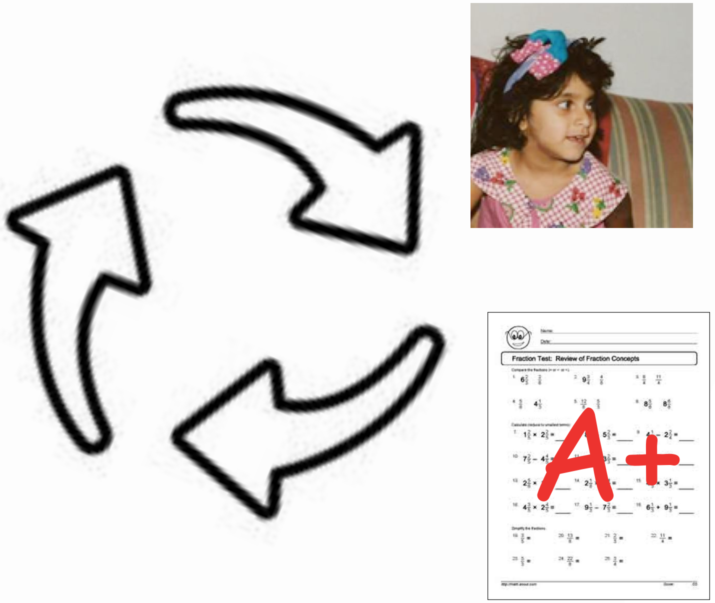
Project Logistics

- **Weekly homework** will build up to the final project (5% of grade)
- Final submission is worth **20% of your grade**
- This project is to be completed **in a group**
  - You will meet with your TA to receive feedback

You MUST iterate based on your TAs feedback

- There are no right or wrong answers to design problems.
  - But there are better and worse answers
- A core skill we want you to learn is to iterate based on feedback. Therefore, your grade is depending on your making your TA happy.





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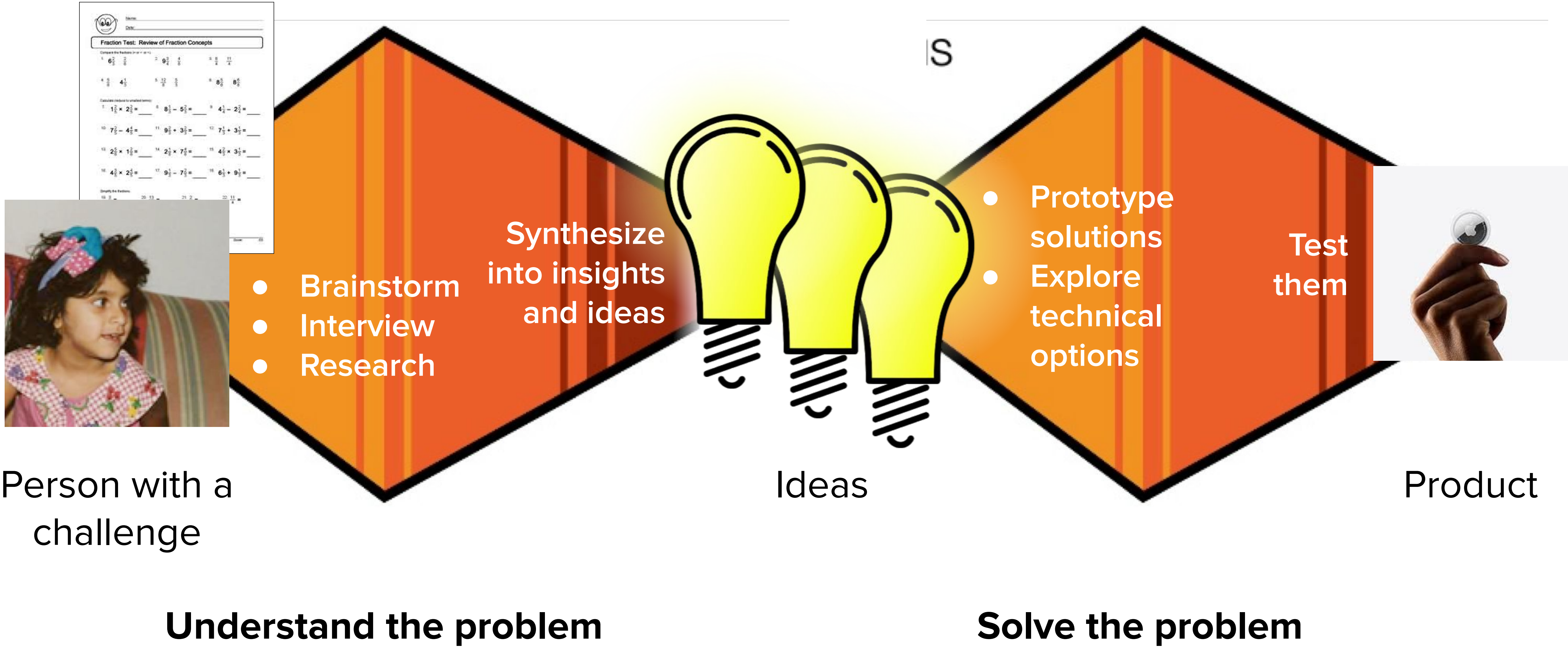
The Iterative Design Process

The biggest misconception about creativity

Idea

Product

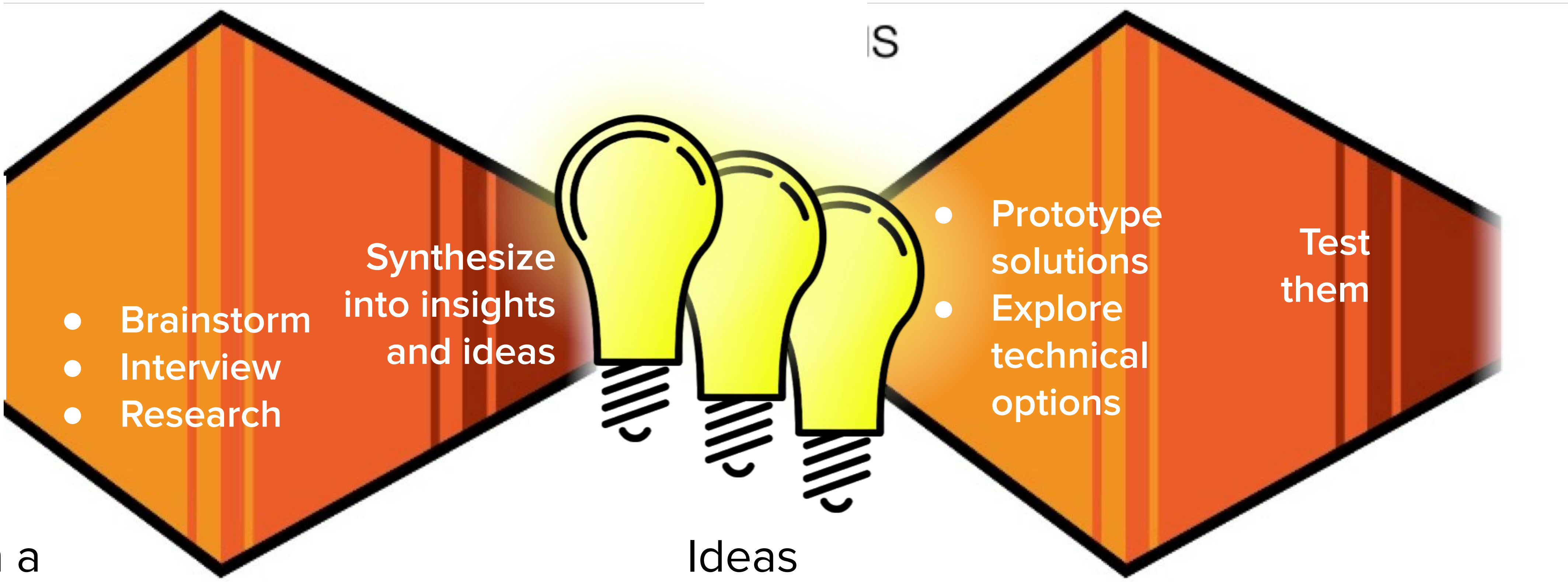
Creativity is a Process



Your USER is your TA (and students in this class). You have to identify a problem you can solve.

YOUR TA

Person with a challenge

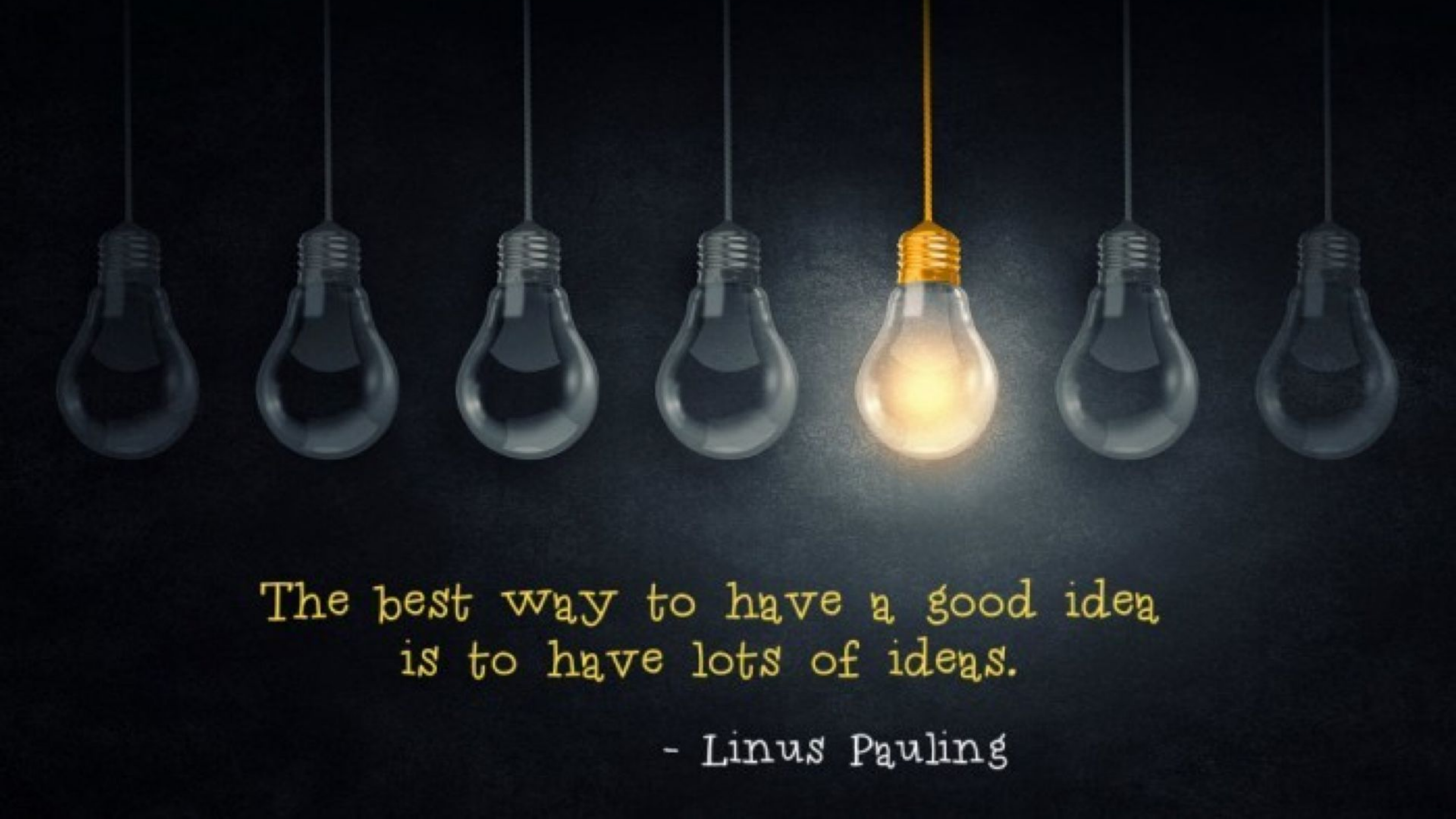


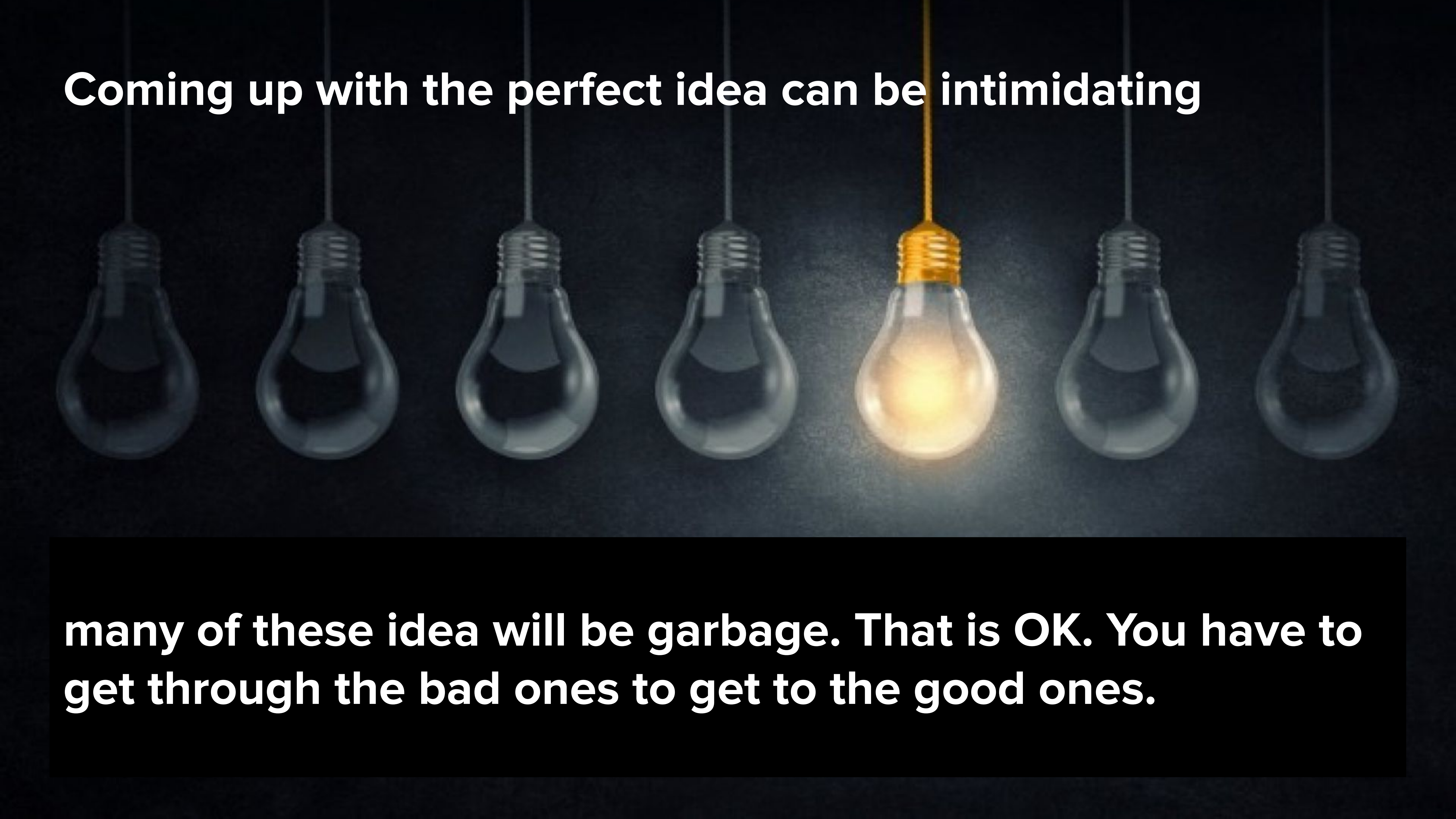
Understand the problem

Solve the problem

HOMEWORK 07 INDIVIDUAL WARM-UP:

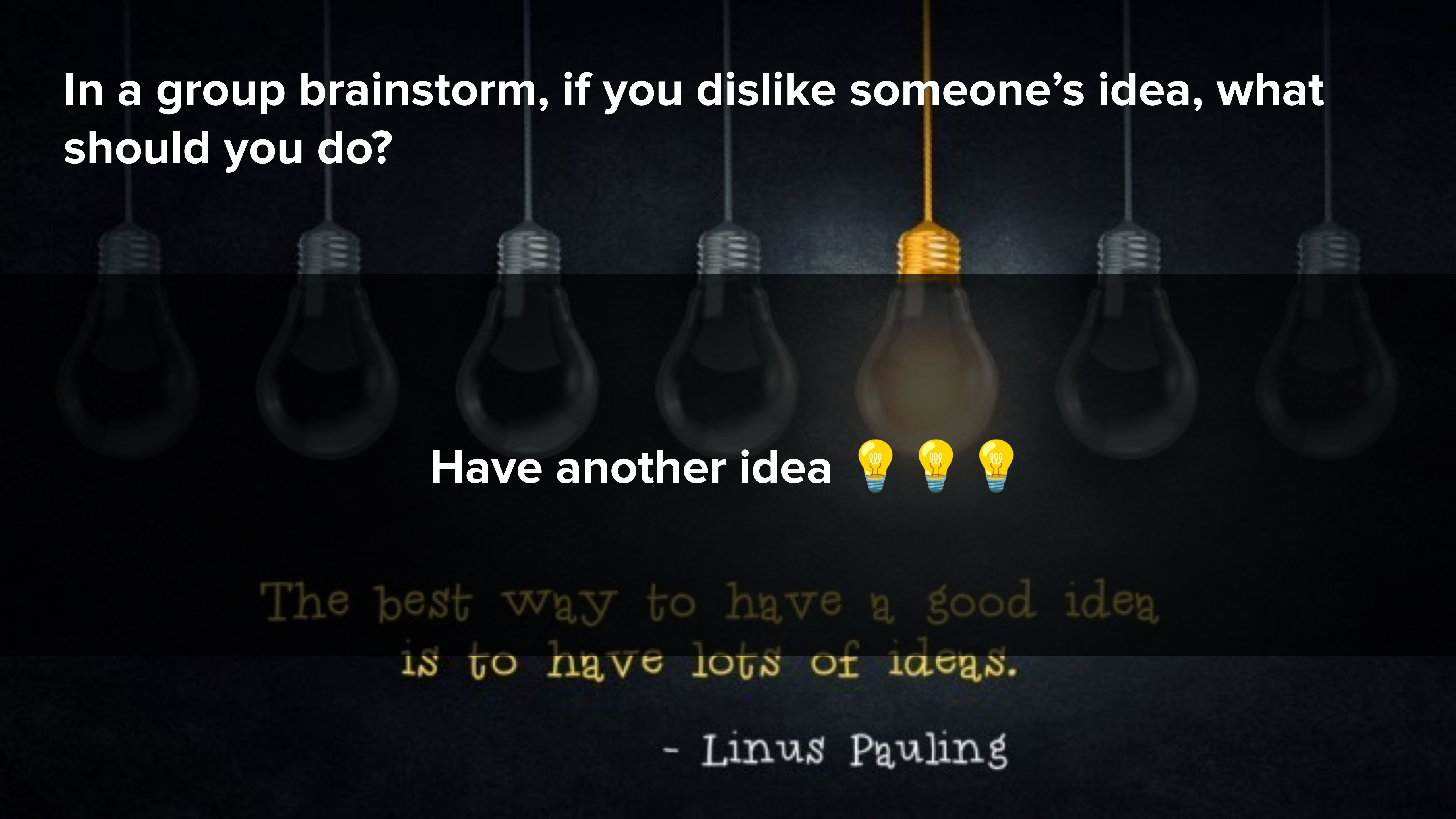
2. **Brainstorm 5 domains**. What are 5 domains that you could teach things to your classmates?
    a. Note: These must be things where you already know a lot of stuff, because you obviously can't teach a topic that you don't know well.
    b. Note: domains are broad like chess, basketball, art history. Next, we'll think about specific topics, which are must less broad
3. **Brainstorm specific topics.** For each of those 5 domains, list 5 specific topics that you could teach interactively in under 10 minutes.
    a. For each specific topic,
        i. What media would you use?
        ii. What would you quiz users on to test their knowledge? (a short sentence or phrase is fine)
    b. Note: It has to be hard enough that people don't already know it, but specific enough that you can teach it in 10 minutes.
    c. Note: To get from a broad domain to a specific topic, you may need to narrow the topic down more and more. For example: Art History → The Modern Era → Identifying types of paintings → identifying post-impressionist paintings.
2. **Select favorite topics.** Of these 25 potential topics, which are your favorites? List between 3 and 6 topics that you would be willing to pursue. PRESENT THESE IN YOUR SLIDES FIRST.


The best way to have a good idea
is to have lots of ideas.
- Linus Pauling


Coming up with the perfect idea can be intimidating
many of these idea will be garbage. That is OK. You have to get through the bad ones to get to the good ones.

Keep the ideas flowing!
Do not be critical of the ideas.

You can prune out the bad ideas later.


In a group brainstorm, if you dislike someone's idea, what should you do?
Have another idea 💡 💡 💡
The best way to have a good idea is to have lots of ideas.
- Linus Pauling

Brainstorming

Domains you could teach

- Cooking
- Programming
- Languages
- Yoga
- Fitness
- Music / music theory
- Games
- How to dance on Tik Tok
- Sit-coms / Cartoons
- Fashion
- Wine
- Cars
- Travel
- Comic Books
- Anime
- Notable architectural buildings
- Astrology
- Photography
- Identify poisonous plants
- Rules of basketball / volleyball
- Flowers
- Art History

Brainstorming

Specific topics to teach cooking

- How to tell when meat is cooked?
- Learn different pasta shapes
- How prepare sashimi
- How to fold dumplings
- Vegan meat substitutes
- How to tell if fruit is fresh

Brainstorming

Specific topics to teach in the dance domain

How to do:

- Tik tok dances
- the Moonwalk
- Fortnite dance
- Stanky leg (specific song)
- Ballet positions (five basic ones, explain what they are and how to add onto them)
- Stretching for dance
- Steps for ballroom dancing

For this class, topics must be small enough that it can be learned in 10 minutes; but also hard enough that the user does not already know it.

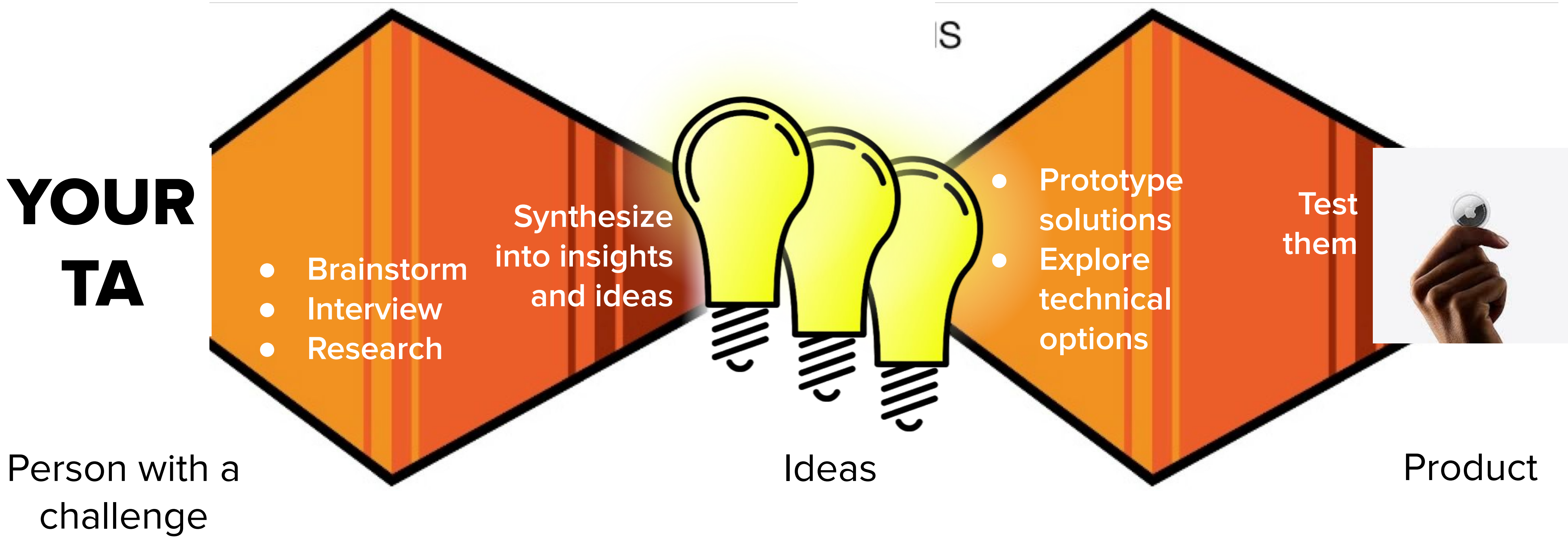
Good Topics

Brainstorm domains and topics…

but we’ll have to narrow them down.

To pick an idea, talk to users and learn about their experience.

To understand the problem, you have to understand your users’ experiences



Understand the problem

Solve the problem

Phase 1: Understand the problem

Talk to Users

If you ask “What are your problems?” you get answers like:

I don’t like eating vegetables

I don’t like speaking up in class

I don’t like doing homework

I don’t like paying taxes

I don’t like my mobile phone carrier

We want to get beyond the obvious,
And get insights about what parts of this they hate.

To identify the users' pain points and *find opportunities to help them*, we need to dig into the details of their experience.

I don't like speaking up in class.

What's the experience of speaking up in class?

Step 1: Find a real person (user) who has done _____ recently

Jamie – a student in User Interface Design who is shy, but is forced to participate in class and fill out a form after class to record her participation

Step 2: Ask the user about the last time they did ____

When was the last time you spoke up in class?

- What did you say?
- Why did you decide to speak up then?
- What did it feel like. Easy? Hard? Scary?
- What happened after you spoke up?
- What did you think/feel/say/do?
- Then what?

Why not ask a broad question: *"Why don't you like speaking up in class?"*

People are better at accurately recalling a specific incident. Asking about a general experience will force them to synthesize and generalize on the spot.

We want to get data on one raw, authentic experience and synthesize it ourselves.

Step 3: Listen to the users' story. Ask questions to help elicit the chain of events. At each stage, ask:

What were they doing?

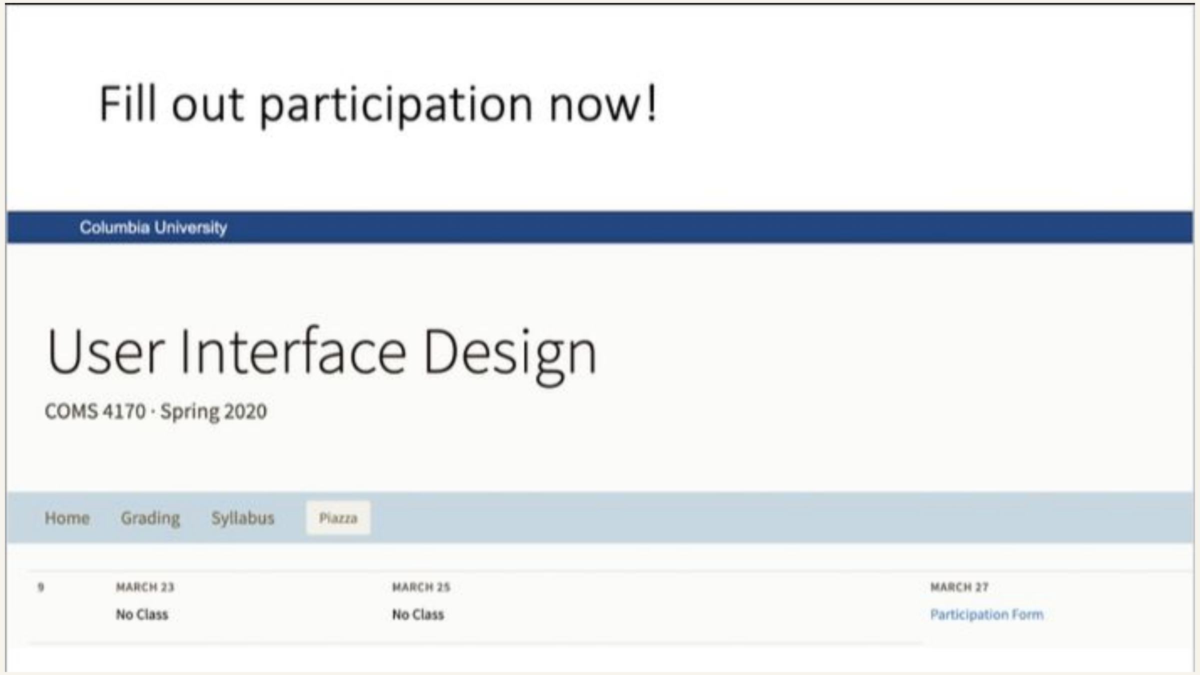
What were they thinking?

What were they saying?

Student answers to: “Tell me about the last time you participated in class?”

I’m worried my accent won’t be understood	I always forget to fill out the participation form
I only saw something if I’m 100% sure of the answer. I don’t like to guess	I always forget to fill out the participation form
I’m so nervous about participating that I don’t pay attention	I always forget to fill out the participation form
It takes me a few seconds to think of something, and by then you’ve called on someone in the front row	I always forget to fill out the participation form

Why is filling out the form so hard?



I remind you at the end of class. Don't you pull out your phone after class anyway?

There is other stuff to check on my phone that I get sucked into.
I'm running to my next class, and my mind switches tasks
I need a computer to fill it out because I have to be logged in
I didn't know there was a deadline to fill them out by!
Can't I just put it on the homework?

Step 4: Synthesize the experience into states. Name them. Map out the emotional highs and lows.

Step 5: Identify insights, opportunities and metrics.

- After class, people want to move on. They're switched mental states.
- **Idea:** Have people just write their participation on the homework.
- Not everybody loves speaking up in front of 200+ people.
- **Idea:** Focus participation on the smaller sections where we can give more feedback.

For Homework 7, interview someone in this class about your domain or topic.

- Interview them about their experience with a domain/topic you are an expert in.
  - What do they already know about the topic?
  - For the domain of “recycling” what do they already know?
  - Where is their knowledge stuck?
- Write down 3 insights you learned. things you didn’t know about the person or problem before hand)
- If they are uninterested in the domain/topic, pick another topic, or another person in the class.

1. **Individual: User Interview.** For one of the domains you are considering teaching, find a target user (someone in this class) and interview them for 10-20 minutes. Get a sense of what they already know about the topic, and what they might be interested in learning about the topic. If you cannot find a user interested in the domain, you’ll have to pick a new domain.
   a. **Interview them.** Turn in a document with 3 insights you got from the interview (things you didn’t know about people’s experiences with this topic and domain before)
      i. Some sample questions to get you started:
      ii. Domain questions: (ex. “Chess” is a domain)
         - What are your experiences around (domain)? If this goes well, you can follow up about more specific topics.
         - What do you already know about (domain)? If this goes well, you can follow up about more specific topics?
         - What’s interesting to you about (domain)?
      iii. Topic questions: (ex. “Opening moves” is a topic in the domain of chess)
         - Have you ever heard of (topic)?
         - What do you know about (topic)?
         - If they don’t know much, you might have to tell them a little bit about it or show an example.
         - If the topic is interesting to them, why is the topic hard for you?
         - If the topic is interesting to them, why? How might they use this knowledge? Ideally, what problem do they have that your website might help them solve.

Interviewing Users

Scenario

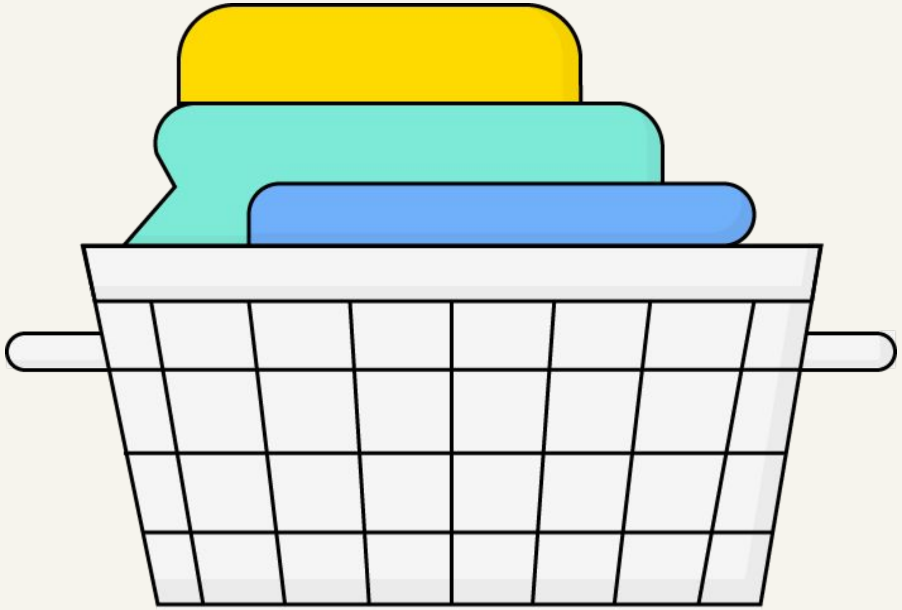
Woes of the city dweller:

- Time
- Unclear pricing
- Quality control

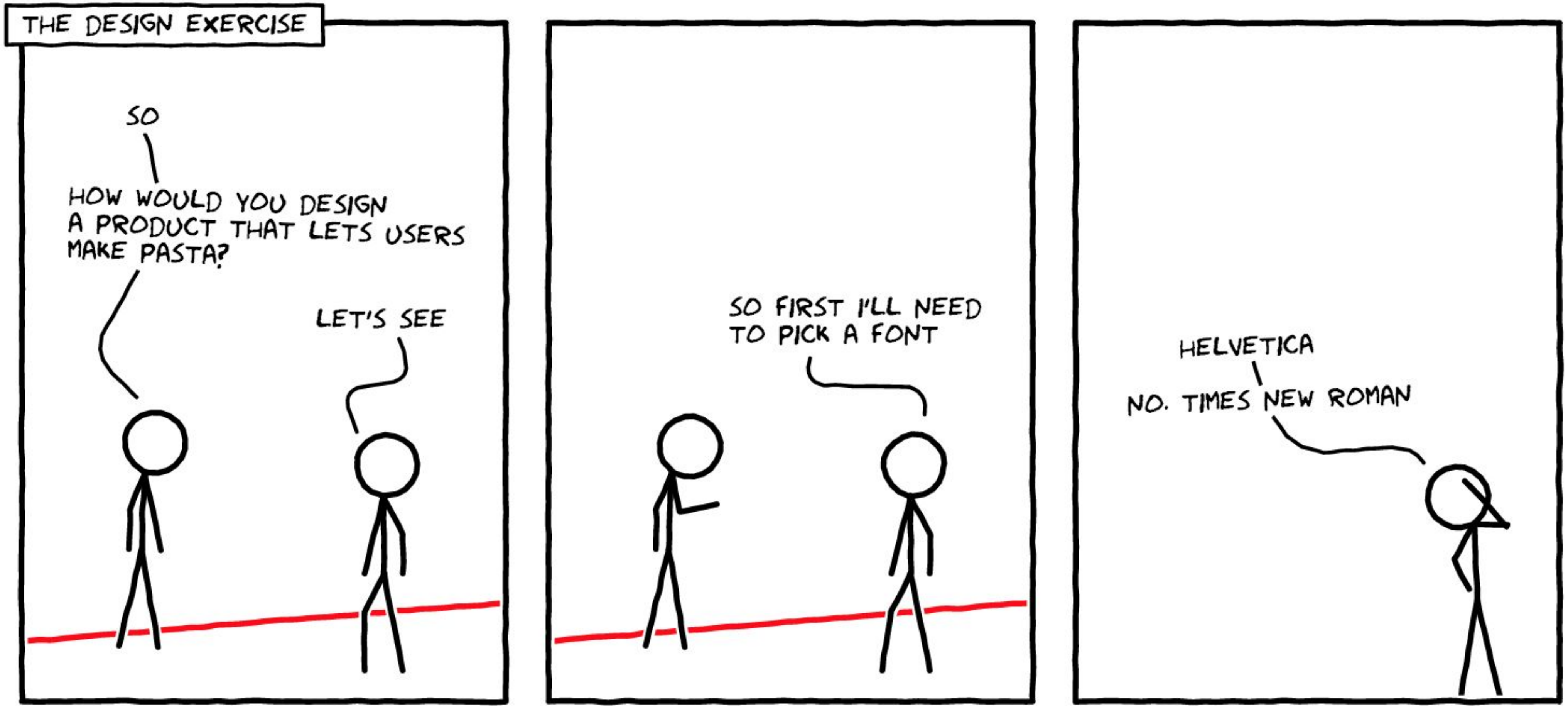
Design Challenge

We've **identified the users pain point**: city dweller who wants to minimize their time at the laundromat by utilizing wash and fold. Now, we're tasked with **finding opportunities to help them**.

Consider opportunities for providing clearer pricing and better quality control.



Where to Start?

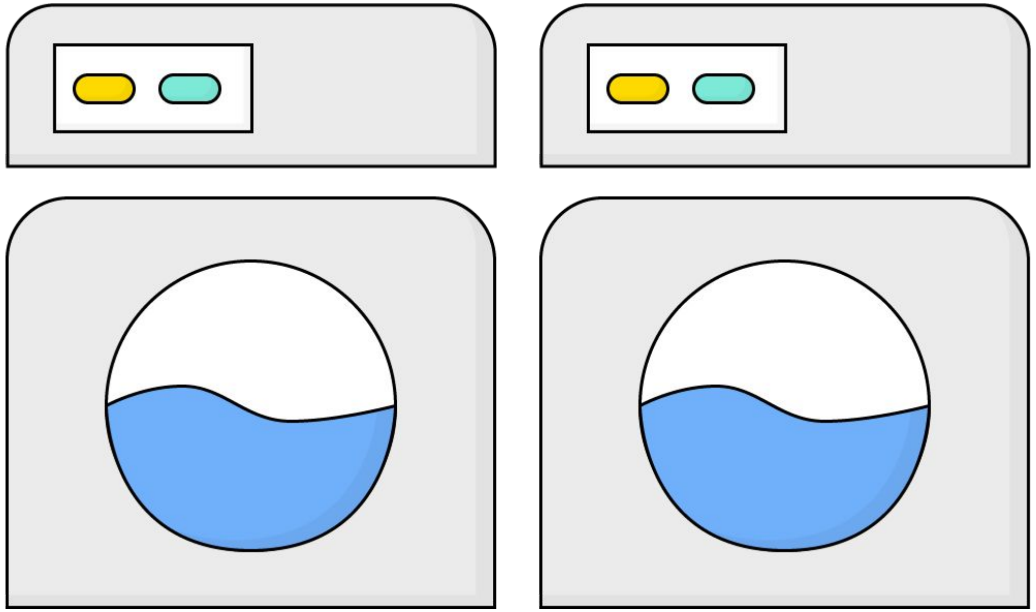


Solo Exercise:

User Research Questions

3 minutes

What are a few questions you'd like to ask potential users about their laundry habits, behaviors, and preferences to get to know their needs?



Which of These Is a Better Question?

Would you like to have your laundry done by using an app?

Describe how you handle your laundry.

Ask open-ended questions.

Which of These Is a Better Question?

Tell me about a time you used wash-and-fold services.

How do you usually use wash-and-fold services?

Focus on specific instances.

Which of These Is a Better Question?

What are your frustrations with wash-and-fold services?

How do you feel about wash-and-fold services?

Avoid leading questions.

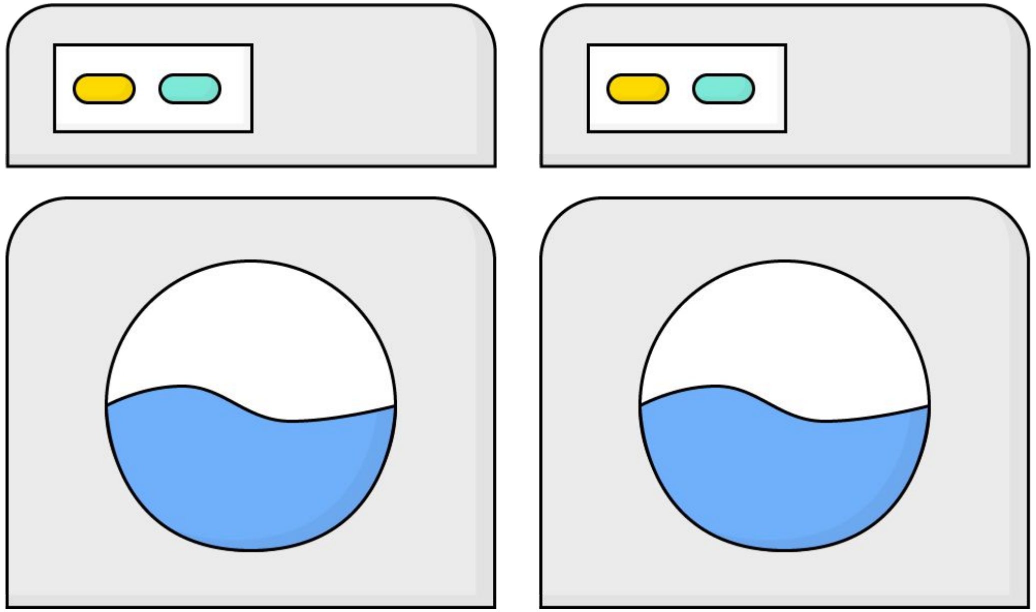
Solo Exercise:

User Research Questions (cont'd)

3 minutes

Now, knowing what we covered here about interview questions, return to what you wrote originally.

How can you make it more powerful, more open-ended, and/or less biased?



Creating an Interview Guide

The interview guide is a document that details what will happen in the interview, your own set of instructions to help carry out the interview.

Consider the following interview outline:

- Introduction
- Initial questions, e.g. “when was the last time you switched phone plans?”
- Main body of the interview
- Wrap up

Reading Ahead Interview Guide

Introduction (5 min)

- *Give out release form and get signature*
- *Turn on video camera*
- *Confirm timing*
- *Who are we and why are we doing this*
- *There no wrong answers, this is information that help us direct our work*

1. We’d like to talk with you today about reading. We have lots of questions to ask you, and we’re interested in hearing your stories and experiences.

Overview (10 min)

2. Can you tell us a little about yourself—what you do, hobbies, etc.?

3. Can you tell me about a recent book you’ve read? Your favorite all-time book?

4. Why do you read?

5. What is your current reading like?

[Probe for different types of reading, locations, motivations, etc.]

6. Is your current reading typical for you? How so/how is it different?

7. Do you call yourself a “reader?” What does that mean to you?

[Look for their categories: could be frequency, importance, etc.]

If you were telling a new acquaintance about yourself, would you talk about reading? What else would you say about yourself?

Exploring Specifics (locations, subject matter, motivations) (10 min)

8. You mentioned (follow up on specifics from overview). Is this always the same, or does it change? Why do you do it this way?

9. Have there been any special circumstances where you’ve done it differently?

Why? How was that?

10. Has anything about the way you do this changed over time? How? Why?

Environment (5 min)

11. What makes a good reading environment for you? What are the elements? What makes an environment not good?

Let’s practice asking each other one interview question

“Tell me about a time you borrowed a book from the library.”

At each stage:

What were you doing?

What were you thinking?

What were you saying?

Now your turn - Try this exercise on a classmate.

“Tell me about a time you borrowed a book from the library.”

At each stage:

What were you doing?

What were you thinking?

What were you saying?

NN/g CUSTOMER JOURNEY MAP TEMPLATE

PERSONA

SCENARIO

USER EXPECTATIONS

	PHASE 1	PHASE 2	PHASE 3	PHASE 4
DOING				
THINKING				
SAYING				

INSIGHTS

INTERNAL OWNERSHIP

CUSTOMER JOURNEY MAP TEMPLATE

	PERSONA	SCENARIO	USER EXPECTATIONS	
	First Name			
	PHASE 1	PHASE 2	PHASE 3	PHASE 4
DOING				
THINKING				
SAYING				
INSIGHTS				
OWNERSHIP				

Switch roles!

“Tell me about a time you borrowed a book from the library.”

At each stage:

What were you doing?

What were you thinking?

What were you saying?

NN/g CUSTOMER JOURNEY MAP TEMPLATE

PERSONA

SCENARIO

USER EXPECTATIONS

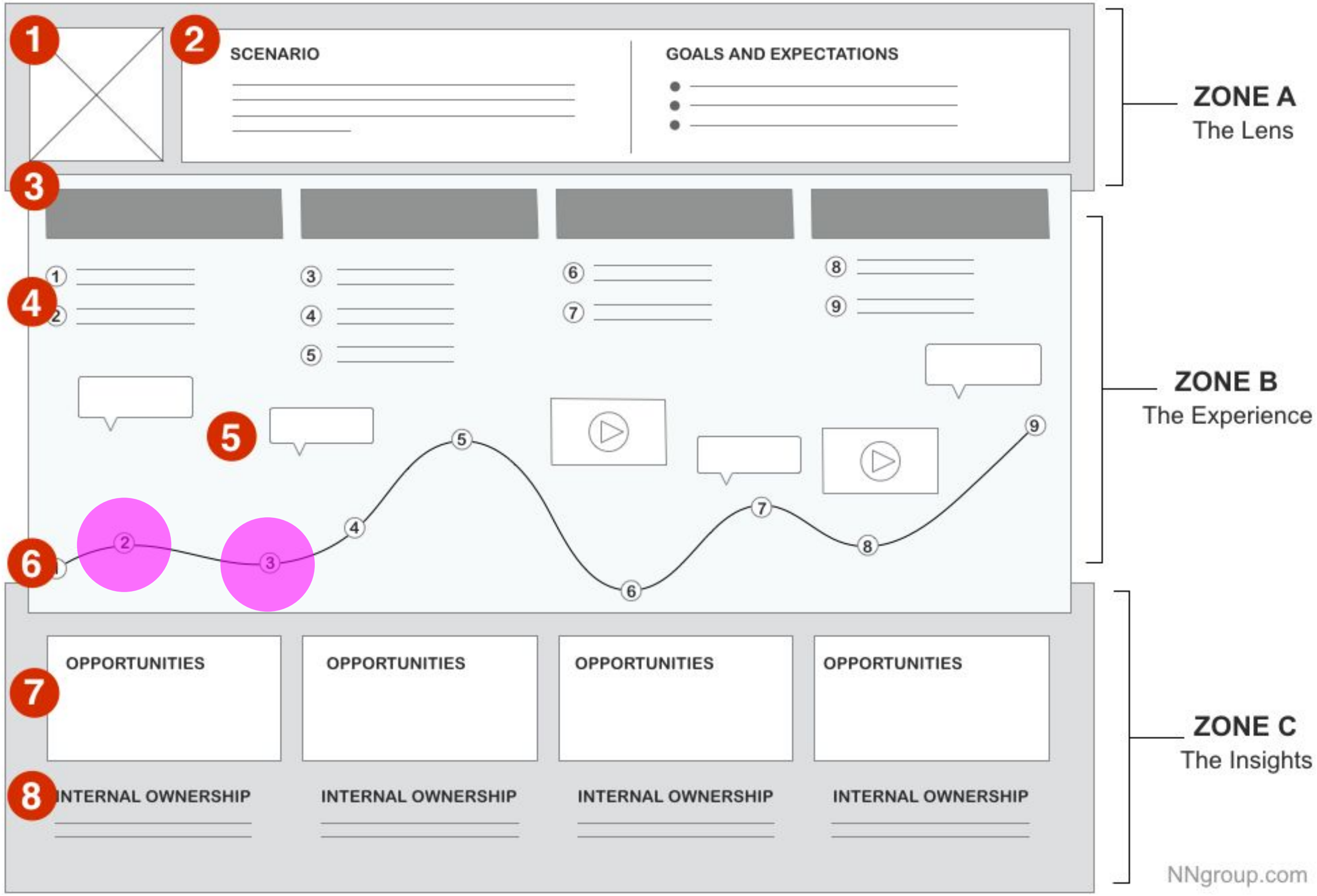
	PHASE 1	PHASE 2	PHASE 3	PHASE 4
DOING				
THINKING				
SAYING				

INSIGHTS

INTERNAL OWNERSHIP

But you’re not ready to design solutions yet.

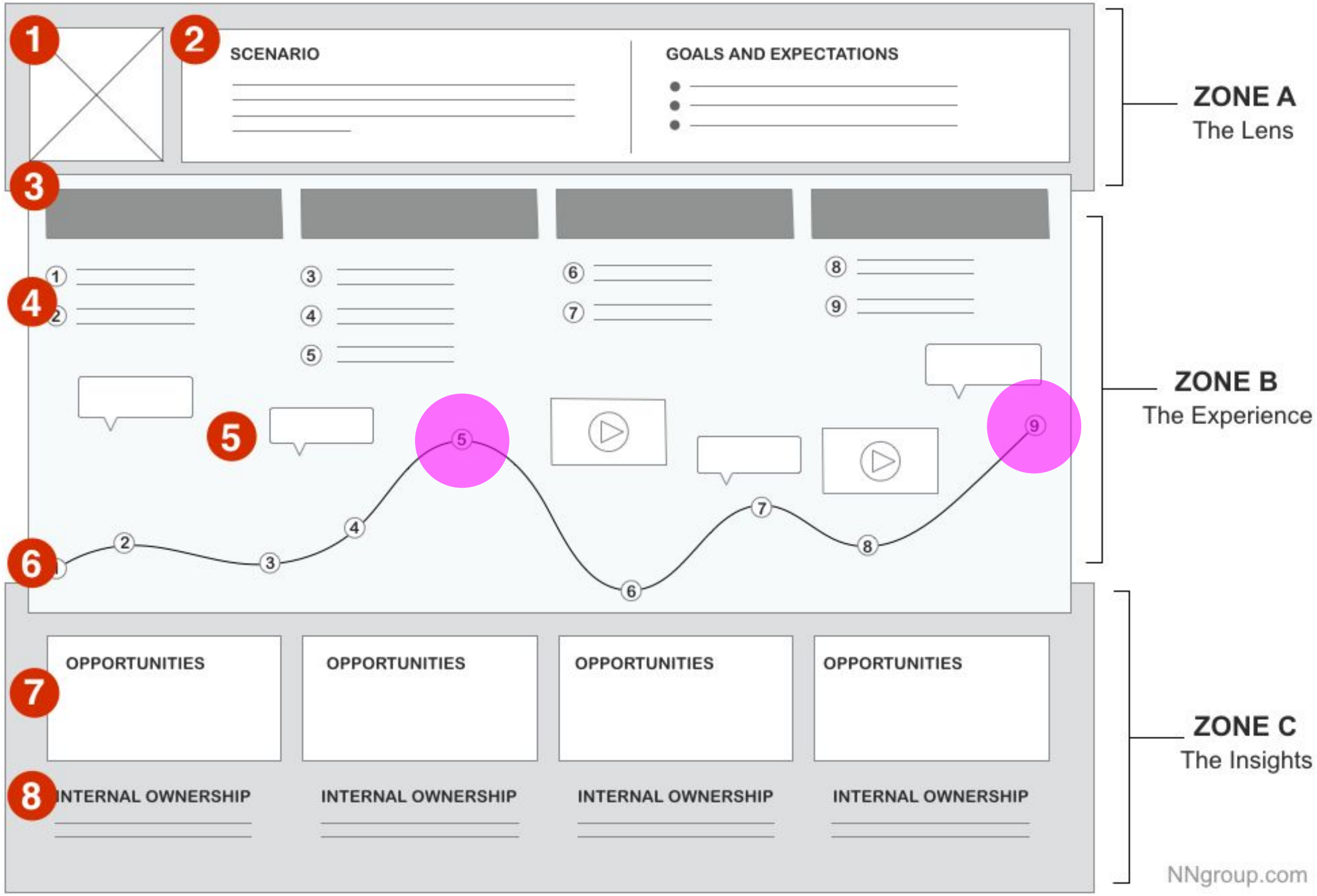
Once you have the map, now what?



Look for **points** in the journey **where expectations are not met**. Users go into an interaction with certain expectations.

When the interaction does not meet their expectations, you see pain points in a customer journey.

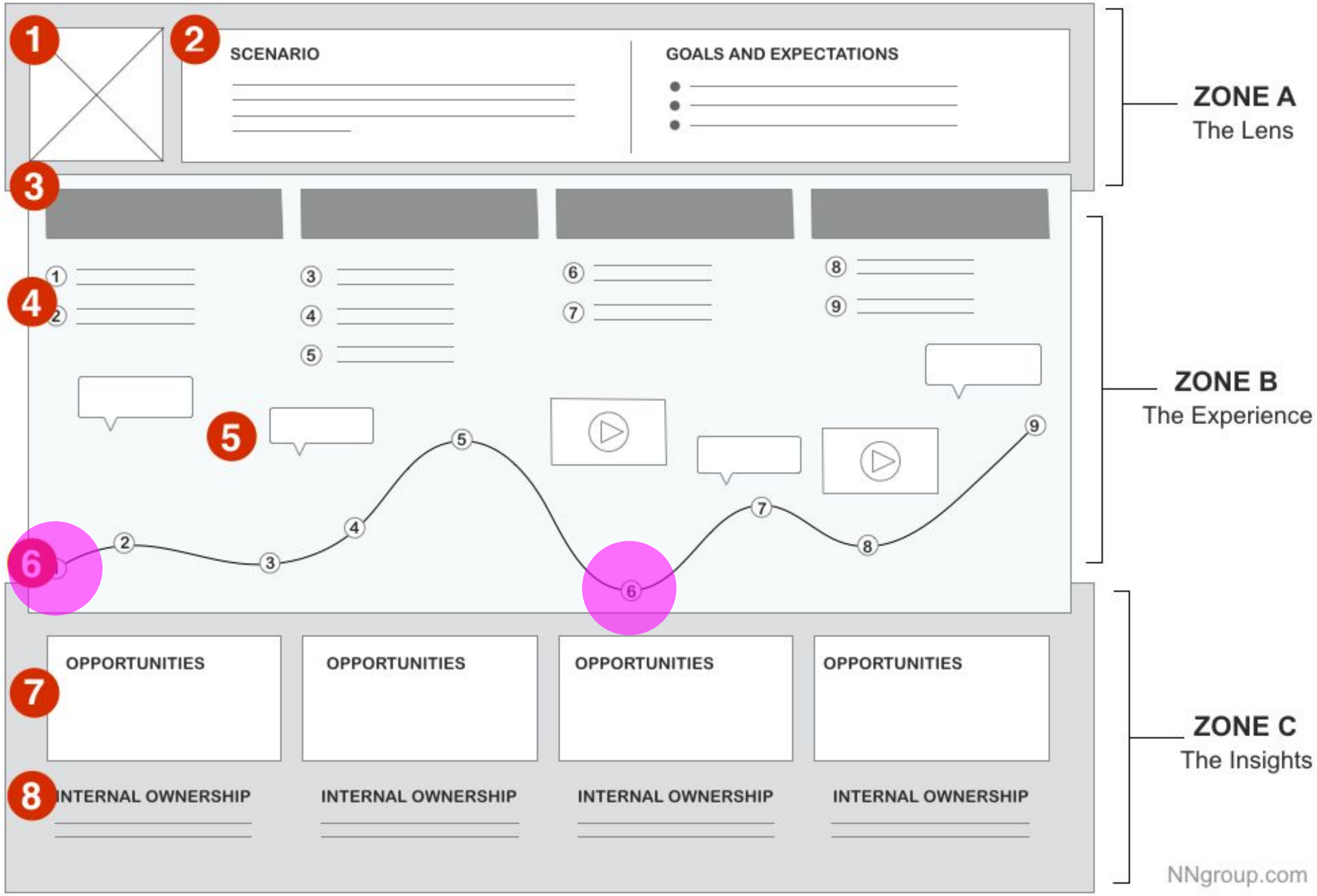
Once you have the map, now what?



Identify **high points** or points where expectations are met or exceeded. Look at the high points in the journey — the interactions that users are happy with.

Where do they express positive thoughts and emotions? These insights are also valuable. You may be able to amplify them or recreate similar experiences elsewhere in the journey.

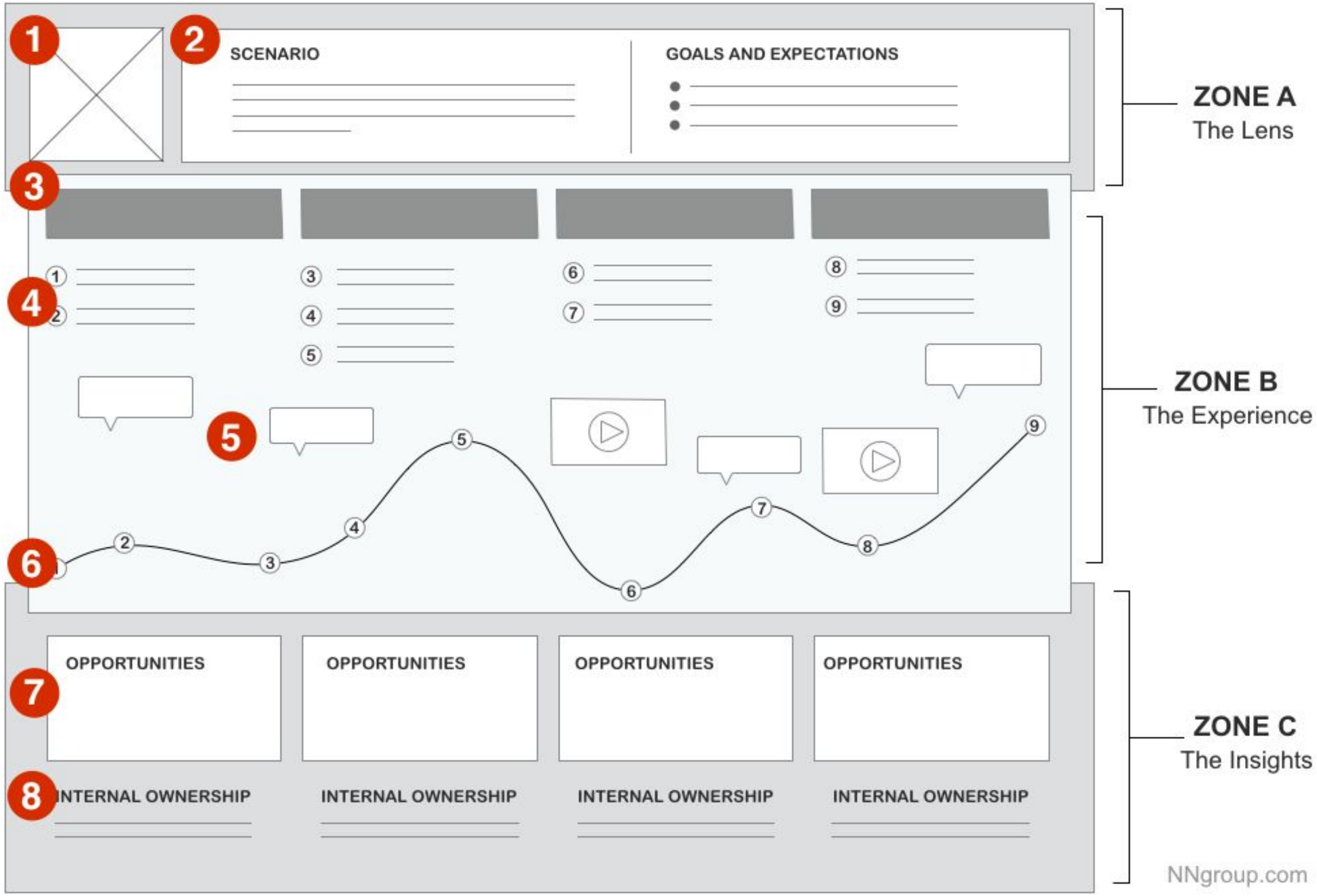
Once you have the map, now what?



Identify the **low points** or points of friction. These points are usually represented visually as dips in the journey diagram.

See where the journey reaches its lowest point and compare it to other low points in the journey. These should be on your shortlist of optimizations.

Once you have the map, now what?



Evaluate time spent.

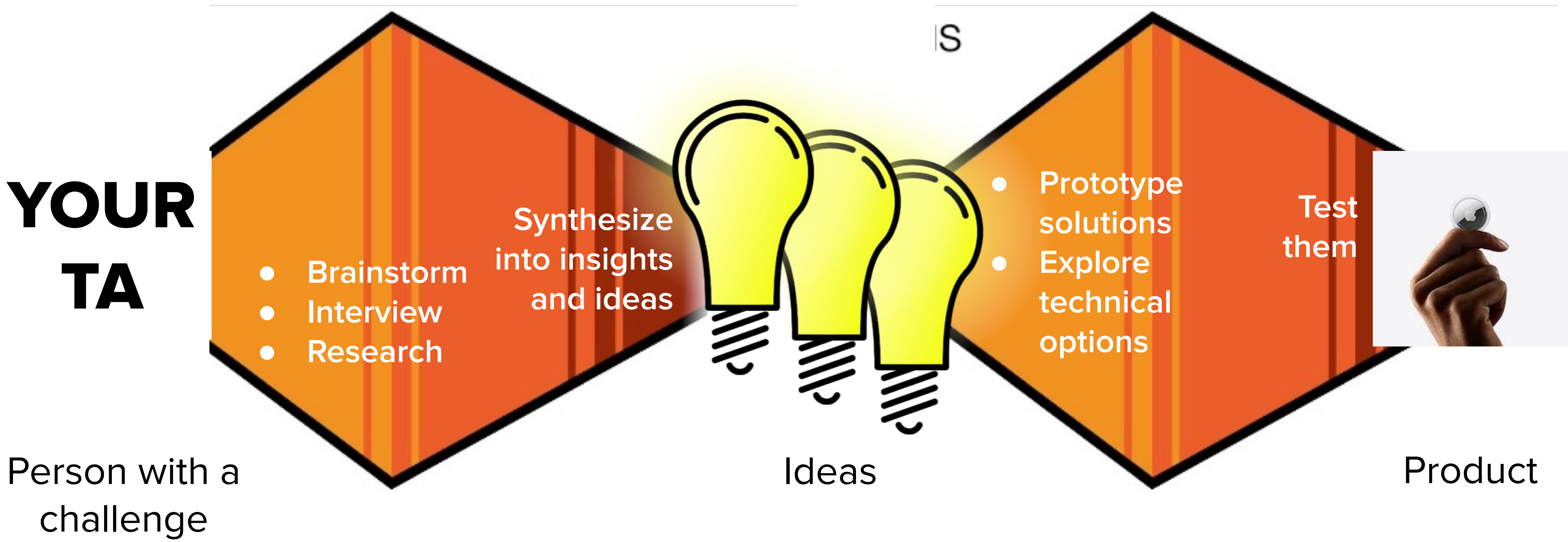
Look for **moments of truth**. This moment may be where your research shows a lot of emotion (positive or negative) or where you see a strong divergence between the paths different users take.

Be sure to look for moments of truth and to call attention to them when you find them.

Phase 1: Understand the problem

Competitor Analysis

To understand the problem, you have to understand your users' experiences



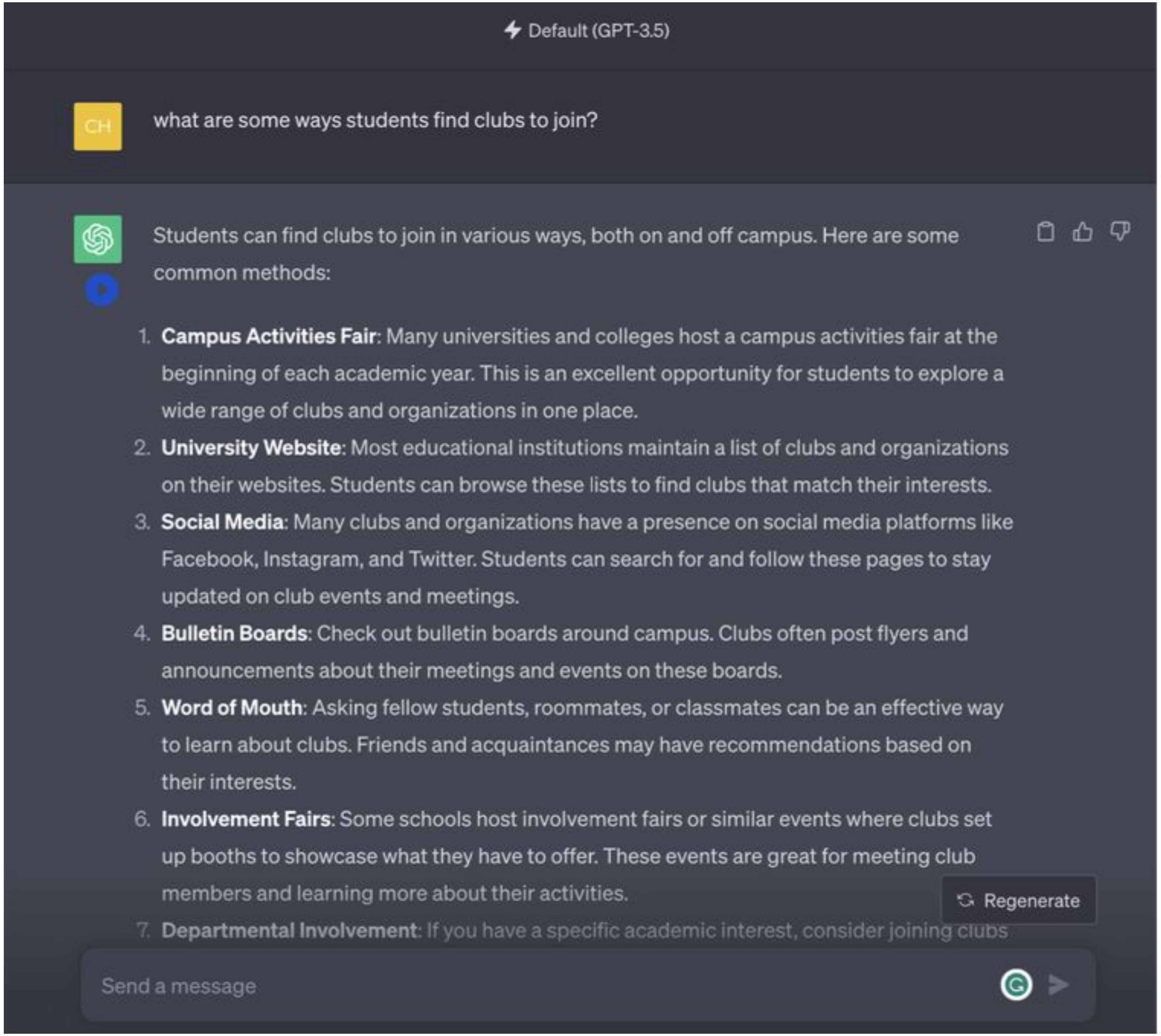
Understand the problem

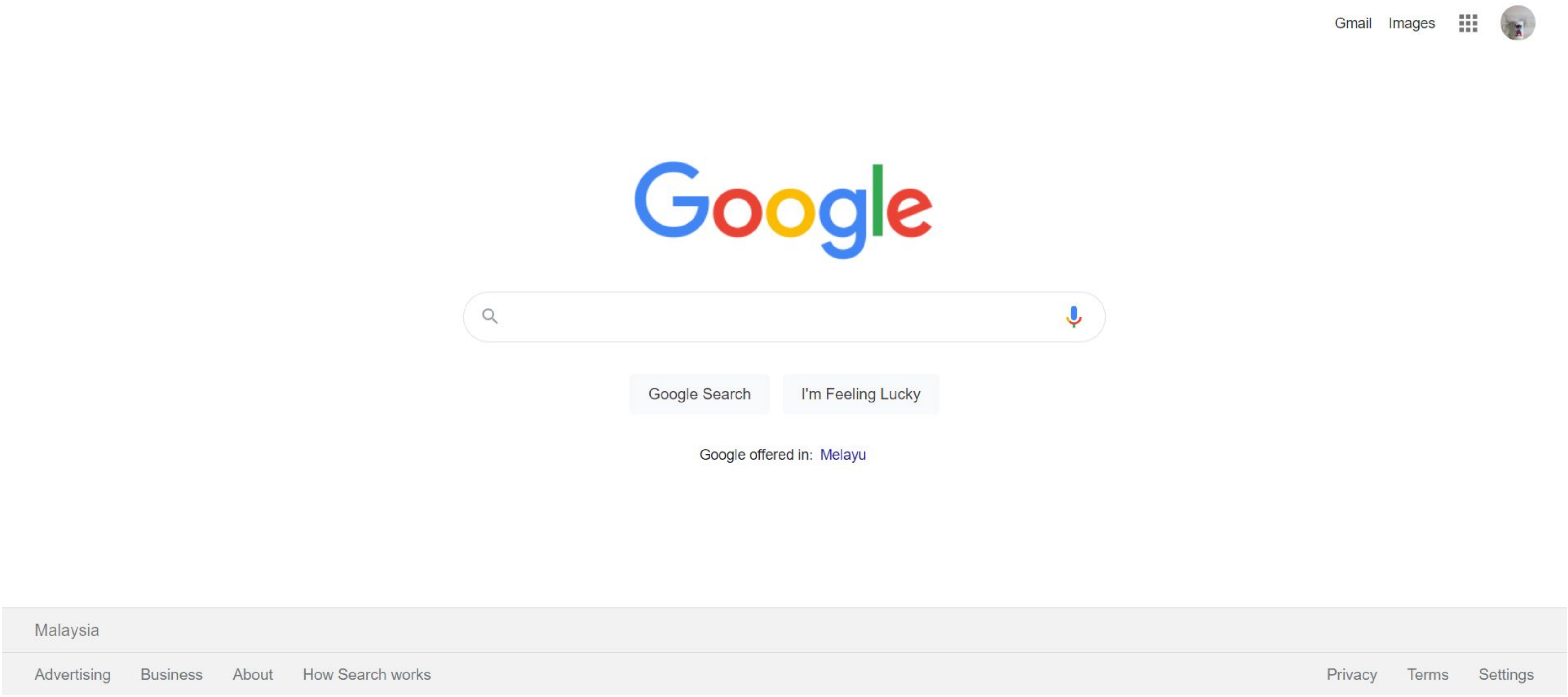
Solve the problem

There are (almost) no new ideas

- Assume if you are thinking about a problem other people have too
- That’s totally fine
- However, it also means you need to do your research

Before you even THINK about solving the problem


Default (GPT-3.5)
CH
what are some ways students find clubs to join?
Students can find clubs to join in various ways, both on and off campus. Here are some common methods:
1. **Campus Activities Fair**: Many universities and colleges host a campus activities fair at the beginning of each academic year. This is an excellent opportunity for students to explore a wide range of clubs and organizations in one place.
2. **University Website**: Most educational institutions maintain a list of clubs and organizations on their websites. Students can browse these lists to find clubs that match their interests.
3. **Social Media**: Many clubs and organizations have a presence on social media platforms like Facebook, Instagram, and Twitter. Students can search for and follow these pages to stay updated on club events and meetings.
4. **Bulletin Boards**: Check out bulletin boards around campus. Clubs often post flyers and announcements about their meetings and events on these boards.
5. **Word of Mouth**: Asking fellow students, roommates, or classmates can be an effective way to learn about clubs. Friends and acquaintances may have recommendations based on their interests.
6. **Involvement Fairs**: Some schools host involvement fairs or similar events where clubs set up booths to showcase what they have to offer. These events are great for meeting club members and learning more about their activities.
7. **Departmental Involvement**: If you have a specific academic interest, consider joining clubs
Regenerate
Send a message


Gmail
Images
Google
Google Search
I'm Feeling Lucky
Google offered in: Melayu
Malaysia
Advertising
Business
About
How Search works
Privacy
Terms
Settings

I am guilty of not asking the internet

- I joined a startup to do online therapy. When I joined, no one had researched what's already out there.
- There were definitely competitors
- I went and tried them all out and learned what they did well and poorly.

This is so obvious it hurts. However, people constantly skip this step. Why?

- Perhaps, people don’t do it because they think of myself as being “original” and “creativity” and “individualistic.”
- I’m excited about building something new. I don’t want there to be a solution out there. So I don’t look for it.

(I don’t know the answer to this. Do you?)

Don’t worry about originality. Worry about solving the problem.

- You’re going to steal ideas from other people
- It’s called background research
- Your unique ideas are probably just a recombination of stuff you’ve seen before, anyway

Design Research

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Allow these ideas to sit with you before jumping to a solution

Problem:
How to better help new and incoming students find clubs and events on campus

What’s the obvious solution?

Direct Competitor Definition

Direct Competitors are the ones who offer the same, or a very similar, set of features to your current or future customers, which means they are solving a similar problem to the one you are trying to solve.

Direct Competitors for finding clubs and events

SEARCHING & BROWSING

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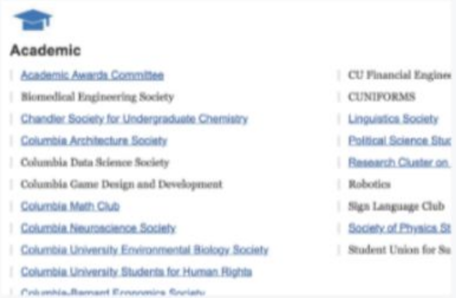

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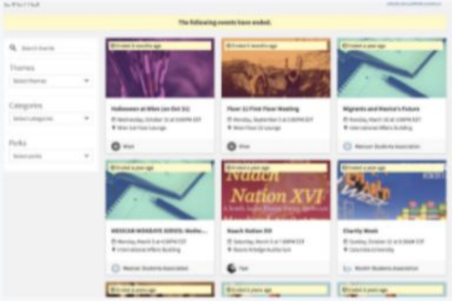

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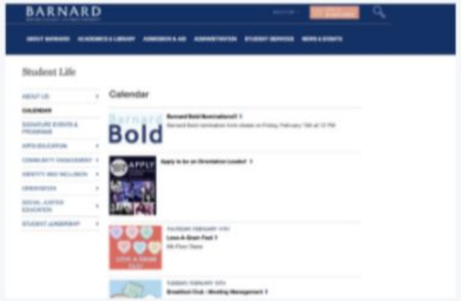

Barnard Clubs & Events

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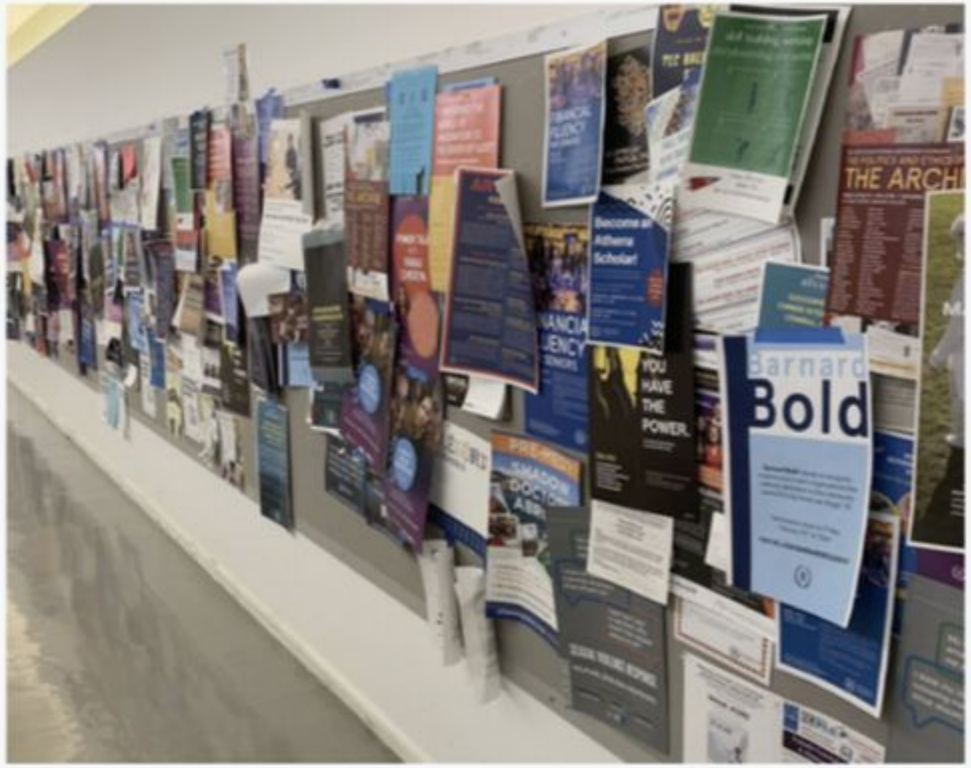
Direct Competitors for finding clubs and events

My dorm floor poster board

Most of the flyers for clubs were for events that happened over a week ago or even last semester. Also, there wasn't equal representation of the 500+ organizations on campus on this one board.



Academic hall poster board

I watched people walk by this wall of posters for 15 minutes, not a single person stopped to look at any of them. It can be an overwhelming experience with too many posters as well.

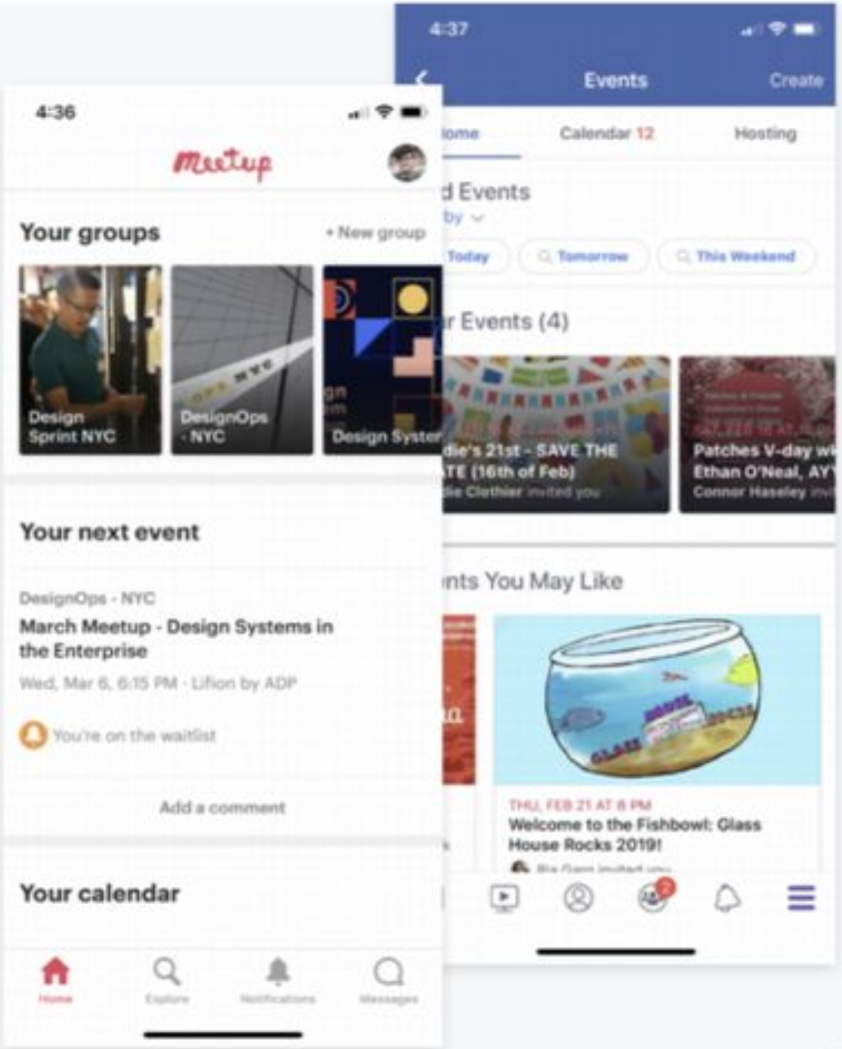
In-Direct Competitor Definition

In-Direct Competitors are the ones who offers a similar set of features but to a different customer segment; which means indirect competitors are solving the similar problem but for a different customer base.

In-Direct Competitors for finding clubs and events

Facebook has a focus around community and personal events, and they make it easy to invite your friends and share information about an event. It is often used as an alternative currently for marketing clubs and event on campus through our Columbia Class pages, but it leaves people not constantly on social media (or without a Facebook) in the dark. Allowing students to still share clubs and events through social media should be a product feature if user adoption is to be successful.

Meetup is a successful event and gathering platform that focuses on specific interests. I've personally had success getting connected to groups around design systems and operations in both Sydney and New York. With Meetup creating interest-based communities around the world, I believe this model could be tweaked applied to campus community as well. Their app encompasses all the functionality needed to meet up efficiently and focuses on the accessibility and visibility of their groups; this focus should be kept for the solution designed for college campuses.



It’s Like X; but for Y

- AirBnB: It’s like Uber, but for hotels.
- Farmers Only: It’s like OK Cupid, but for Farmers (only)
- LinkedIn: It’s like Facebook, but for work associates

A great way to find a solution is to reuse the structure of another solution, but for a different problem or market segment.

You are never the first person to think of this problem

- Newton and Leibnitz invented calculus at the same time
- The steam engine was invented 4 or 5 recorded times before it started the Industrial Revolution
- Alan Turing was the second person to solve the Halting Problem (his advisor Alonzo Church beat him to it)
- Long before Facebook there was MySpace, and Friendster and a bazillion other social networks
- DeepSeek and Kimi.ai released deep reasoning models within a day of each other

All you want is a solution to a problem. If someone else has already done it, that's great. If they've come close but haven't exactly solved our problem, we want to **understand what they've already done** and how to make it better or different for out problem different problem or market segment.

Design Research

- Identifying the **existing or obvious** solutions
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Allow these ideas to sit with you before jumping to a solution

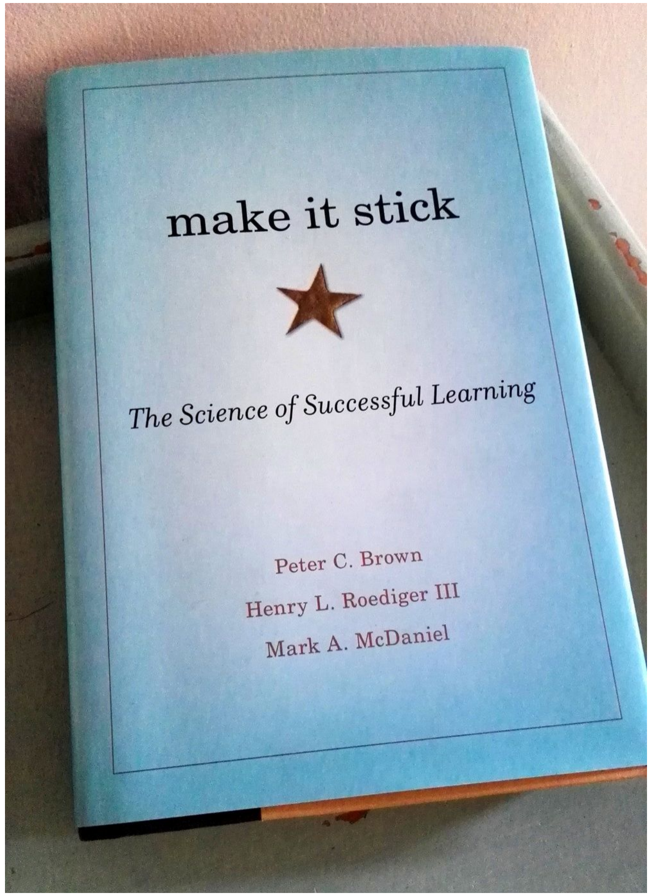
Homework

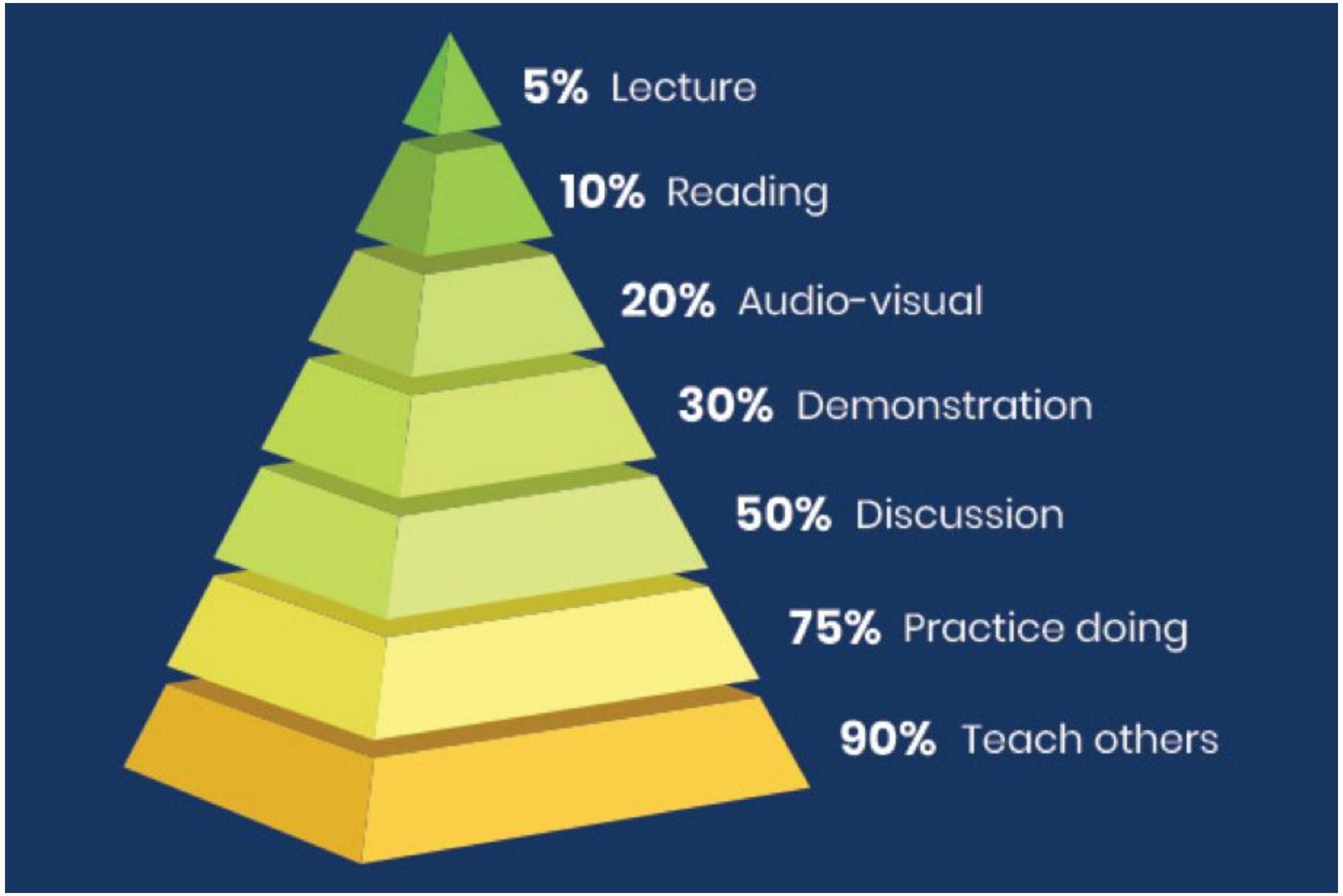
2. **Individual: Competitor Analysis.** In your group, each person should analyze a different competitor product. For your chosen product, do the following:
    a. What topic does it teach? (if it teaches multiple topics – like multiple coding languages), pick one to focus (like JavaScript) on for the sake of concreteness.
    b. Who is the target audience (or who do you think benefits the most from it)?
    c. What media does it use to help people learn? (show a screenshot and describe it in a sentence)
    d. What is a major way it uses interaction to help people learn? (show a screenshot and describe it in a sentence)
    e. What are 3 things you like about the interaction and media usage that help people learn? (what might inspire your design?)
        i. Show a screen shot for each one.
    f. What are 3 things that could be better or different about the interaction and media usage that help people learn?
        i. Show a screen shot when applicable.

Phase 1: Understand the problem

Academic Research

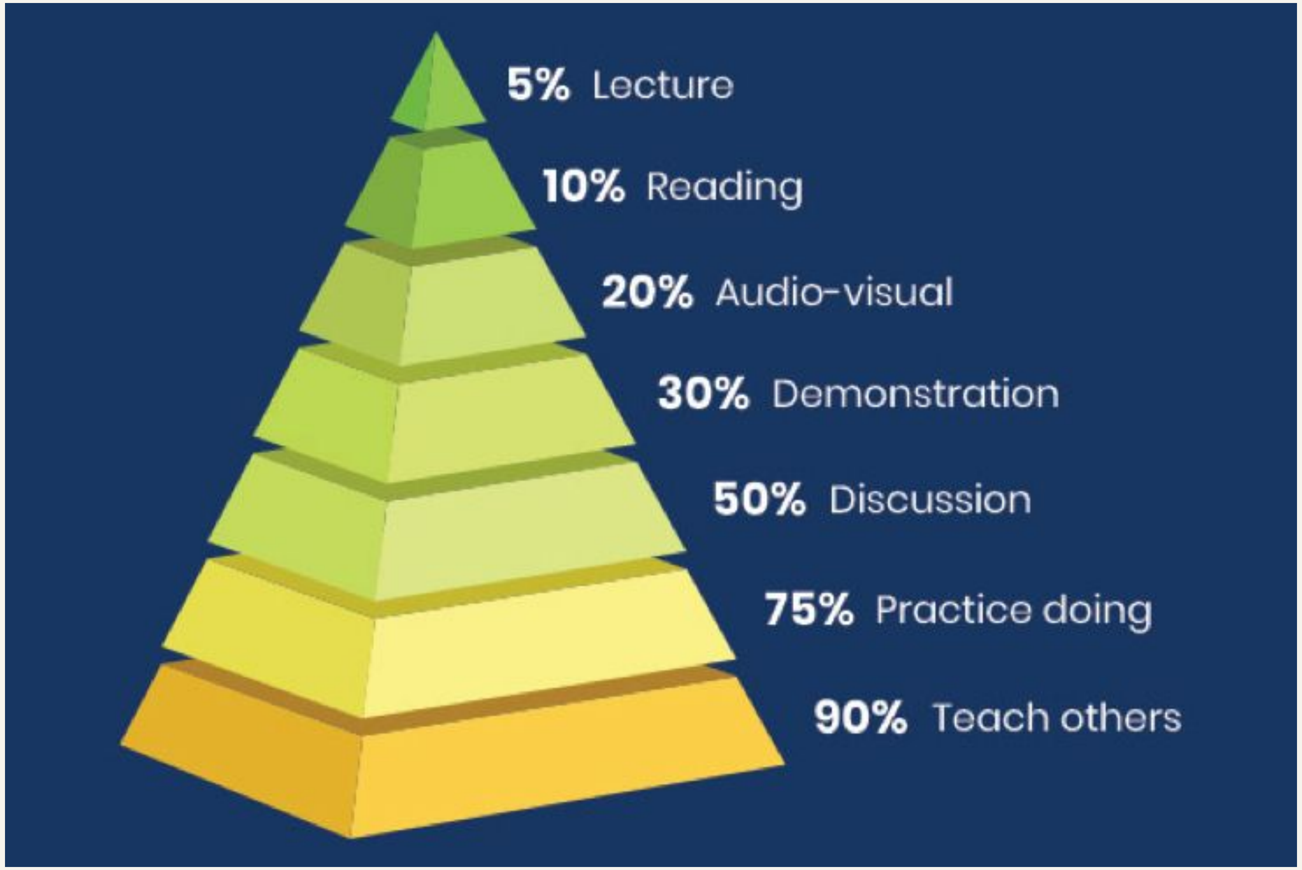
Read books, papers, theories, scientific evidence to get insights into the problem.





Educational Insight #1

Reading textbooks is boring.
Nobody learns from that.
People learn by practicing –
from doing something and
getting feedback.



Educational Insight #1

Students are terrible at assessing their learning. They need tools to assess themselves.

Insights and opportunities can come from anywhere. Just observe and ask why.

In academic studies

Why didn't Aristotle try any of his theories like Galileo did?

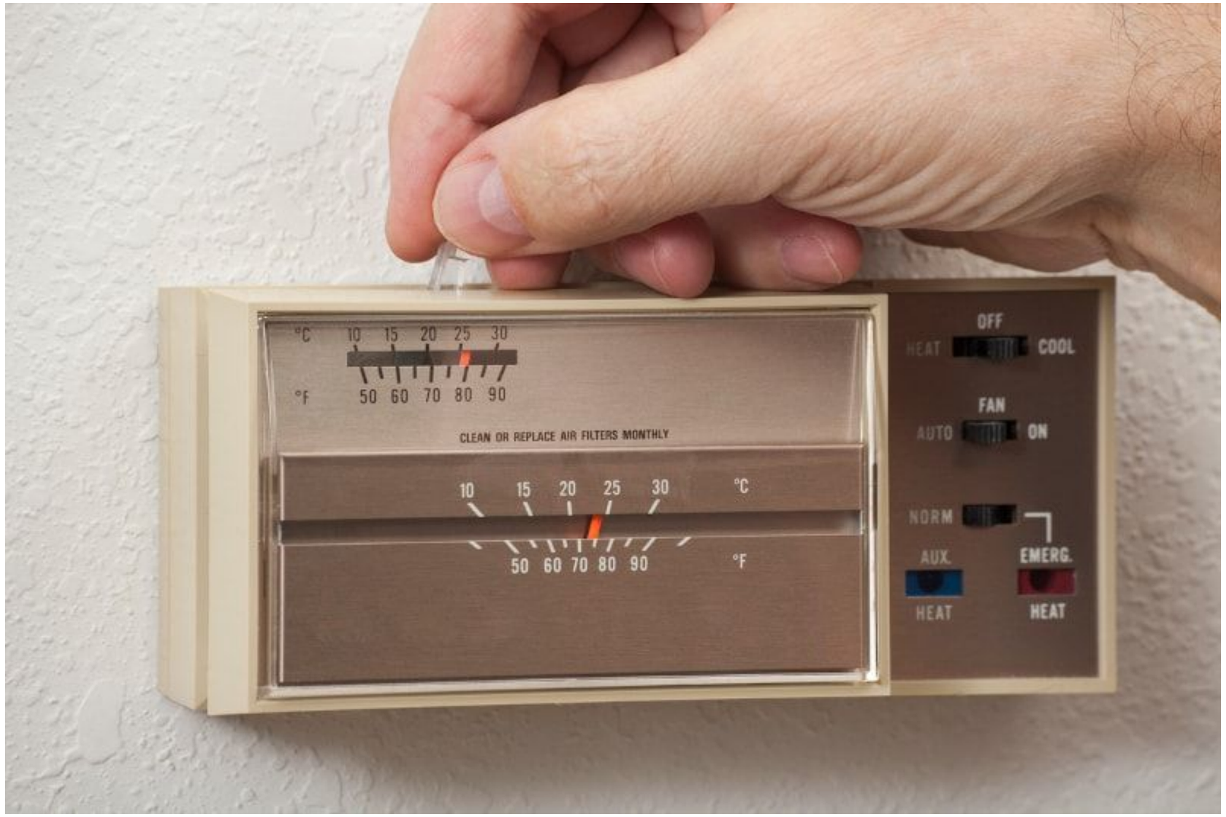
During everyday activities

How do baristas manage to serve all these people?

In the things you find fun

How does Popovic always construct a playoff team?

Everyday activity: changing the thermostat



Why do I constantly have to set and adjust this thing?

Real insights behind applications you use.

Problem	Idea
Teaching fractions	Workbooks suck. I’m going to show people how I think through them problem
Making yearbooks	Photoshop is HARD, expensive, and sharing resources is annoying. Maybe some online templates can make this easier
Social network for photos	People take crappy photos and are not too eager to share them. What if filters made every photo beautiful?

Lecture 01: 10 Usability Heuristics

Homework 07

Homework 7 has 3 parts

- Individual Warm Up due **Monday at 11:59PM**
  - Brainstorm domains and topics you could teach

- Group Warm Up due **Friday at 11:59PM**
  - Groups must be formed by **Friday at 11:59PM**

- Main Homework due **Monday at 11:59PM**
  - User Interview
  - Competitor Analysis
  - Group Presentation: what 5 topics is your group considering?
- Lecture on **Tuesday at 7:00PM**
- TA Session on **Tuesday *(after lecture)***
  - Present topics and get TA feedback
  - Group formation starts

10% of your grade is participation in your TA section

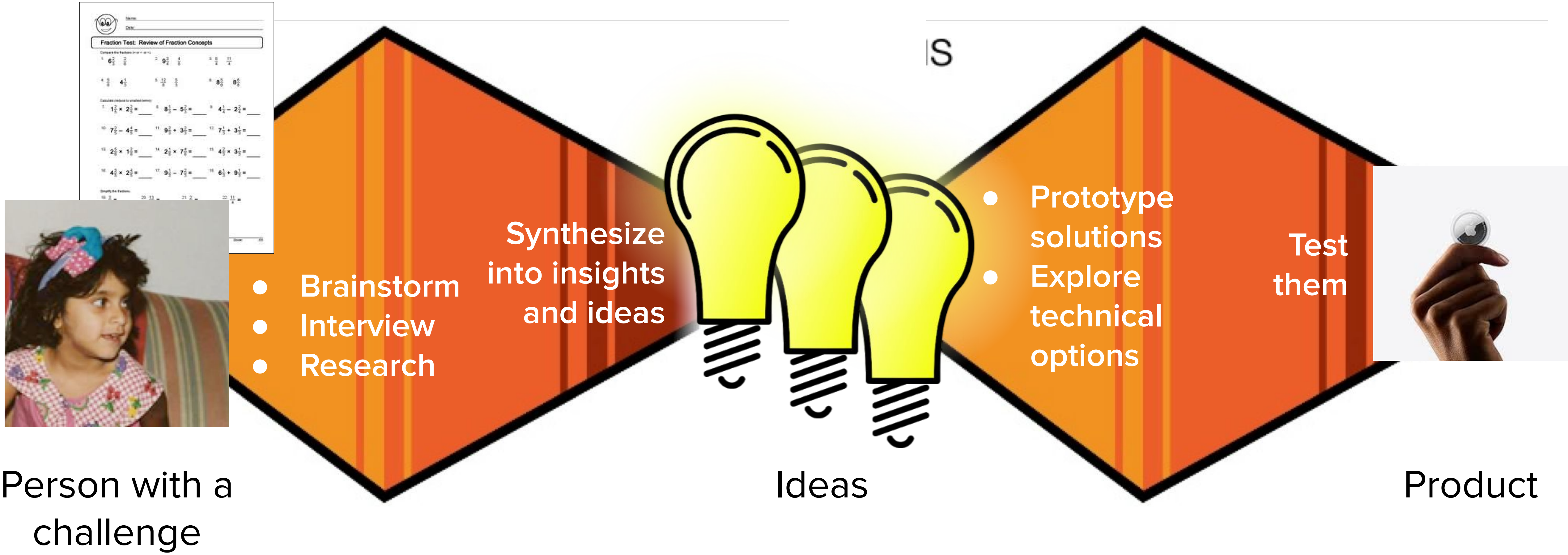
- Show up on time
- Be prepared to present
- Present your materials clearly and concisely.
- Actively, listen to other presentations.
- If you’re virtual, keep your camera on at all times. No exceptions.
- Take feedback from your TA.
- If you can’t attend a feedback session, email your TA in advance to schedule a makeup session before **Friday 11:59PM**

Lesson 15: Brainstorming

Summary

The biggest misconception about creativity

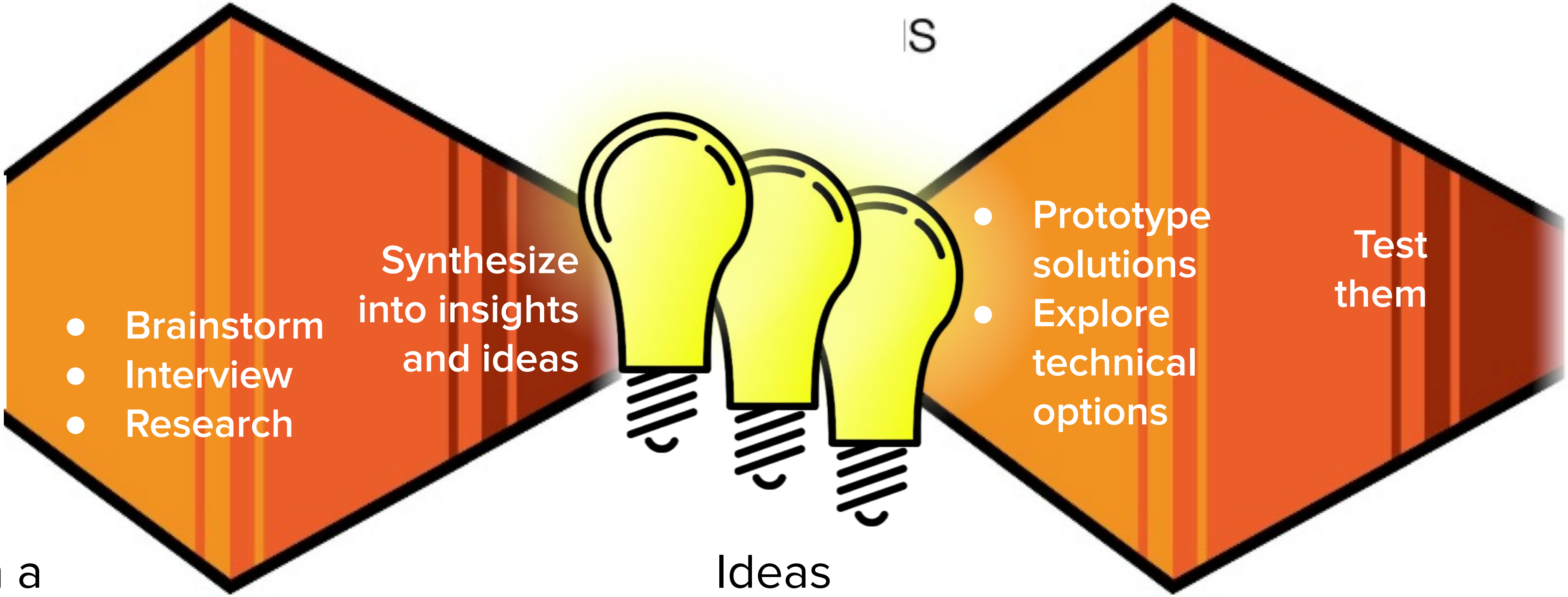
Creativity is a Process



Understand the problem **Solve the problem**

Your USER is your TA (and students in this class). You have to identify a problem you can solve.

YOUR TA

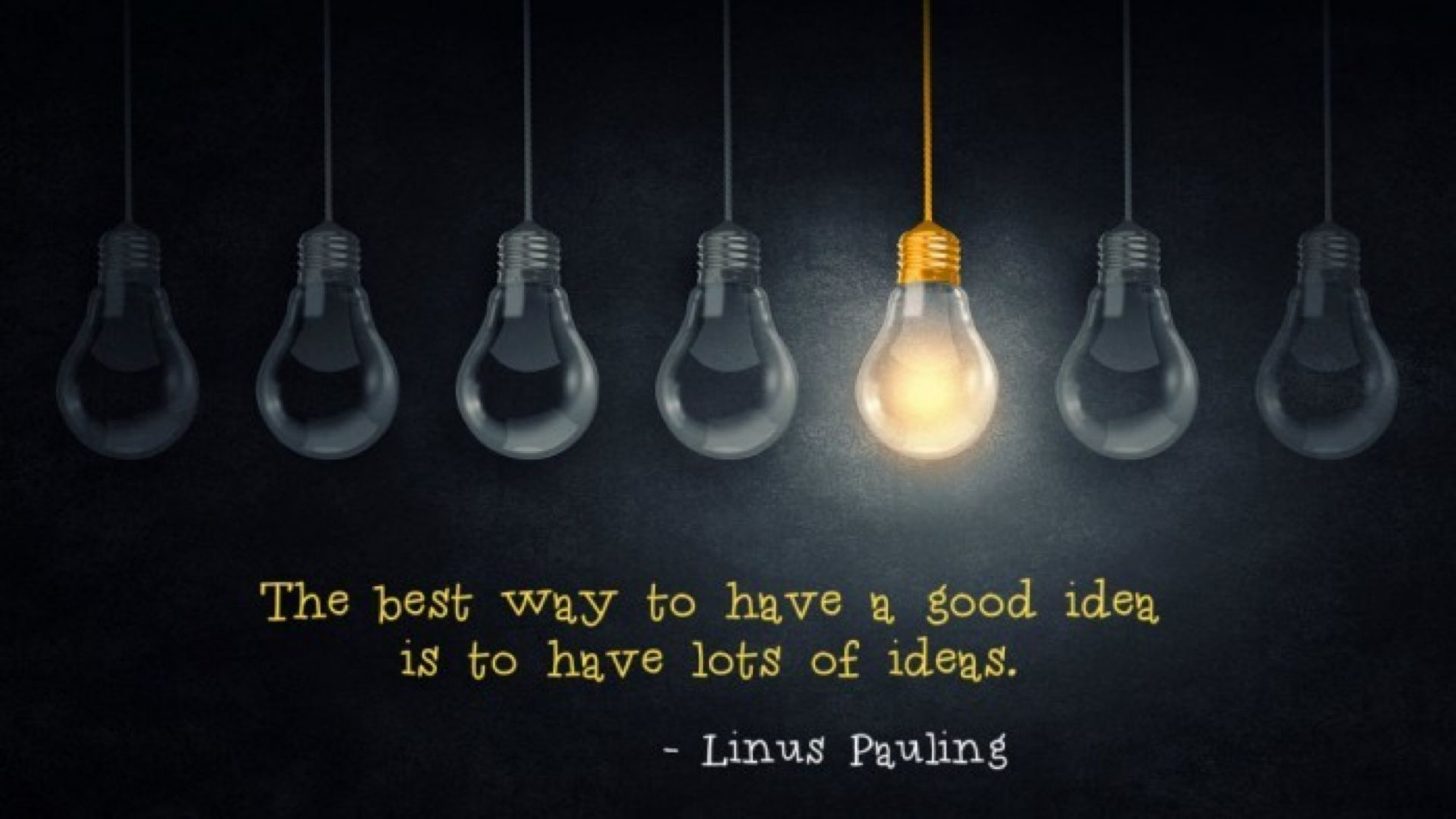


Person with a challenge

Ideas

Understand the problem

Solve the problem


The best way to have a good idea
is to have lots of ideas.
- Linus Pauling

Step 2: Ask the user about the last time they did ____

When was the last time you **spoke up in class**?

- What did you say?
- Why did you decide to speak up then?
- What did it feel like. Easy? Hard? Scary?
- What happened after you spoke up?
- What did you think/feel/say/do?
- Then what?

Research other ways of solving similar problems

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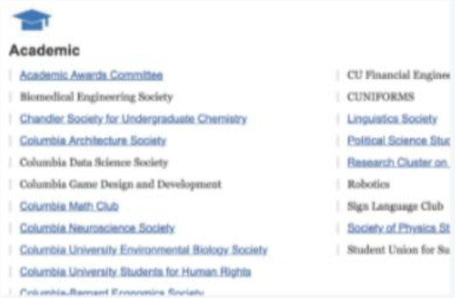

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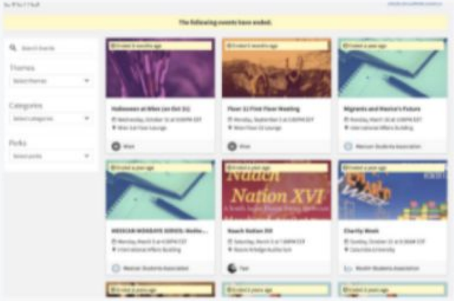

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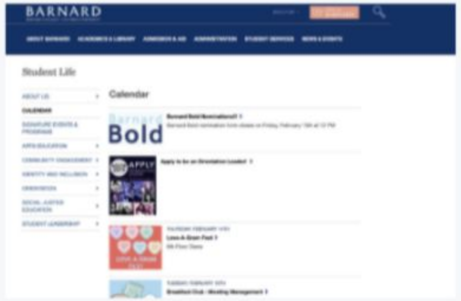

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COMS 4170 · SPRING 2026

Fill out today's Participation Form

Prof. Celeste Layne
Monday March 24, 2026